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INTERNATIONAL DEVELOPMENT ASSOCIATION

PROJECT APPRAISAL DOCUMENT

ON A PROPOSED CREDIT

IN THE AMOUNT OF SDR 203.3 MILLION
(US\$270 MILLION EQUIVALENT
OF WHICH US\$150 MILLION FROM THE CRISIS RESPONSE WINDOW)

TO THE

PEOPLE'S REPUBLIC OF BANGLADESH

FOR A

BANGLADESH SUSTAINABLE RECOVERY, EMERGENCY PREPAREDNESS AND RESPONSE PROJECT

APRIL 30, 2025

Urban, Resilience and Land Global Practice
South Asia Region

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CURRENCY EQUIVALENTS

(Exchange Rate Effective March 31, 2025)

Currency Unit = BDT

BDT 121.50 = US\$1

US\$1.33 = SDR 1

FISCAL YEAR

July 1 - June 31

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ABBREVIATIONS AND ACRONYMS

AM	Accountability Mechanism
B-STRONG	Bangladesh Sustainable Recovery, Emergency Preparedness and Response Project
BADC	Bangladesh Agricultural Development Corporation
BDP	Bangladesh Delta Plan
BNBC	Bangladesh National Building Code
BWDB	Bangladesh Water Development Board
CCDR	Bangladesh Country Climate and Development Report
CCS	Cluster Community Societies
CDD	Community-Driven Development
CEIP-I	Coastal Embankment Improvement Project Phase 1
COM	Community Operations Manual
CPF	Country Partnership Framework
CRI	Corporate Result Indicator
CRW	Crisis Response Window
CSA	Climate-Smart Agriculture
DA	Designated Account
DAE	Department of Agricultural Extension
DDM	Department of Disaster Management
DPs	Development Partners
DRM	Disaster Risk Management
EGPP+	Employment Generation Program for the Poorest PLUS
ES	Environmental and Social
ESF	Environmental and Social Framework
ESMF	Environmental and Social Management Framework
ESS	Environmental and Social Standards
FAO	Food and Agriculture Organization
FAPAD	Foreign Aided Project Audit Directorate
FID	Financial Institutions Division
FM	Financial Management
FPMU	Food Planning and Monitoring Unit
FSCD	Fire Service and Civil Defence
FY	Fiscal Year
GHG	Greenhouse Gas
GoB	Government of Bangladesh
GRADE	Global Rapid Post-Disaster Damage Estimation
GRM	Grievance Redress Mechanism
GRS	Grievance Redress Service
HRP	Humanitarian Response Plan
IA	Implementing Agency
IEG	Independent Evaluation Group

IGAs	Income Generation Activities
iBAS++	Integrated Budgetary and Accounting System
INSARAG	International Search and Rescue Advisory Group
IPF	Investment Project Financing
IRR	Internal Rate of Return
LGD	Local Government Division
LGED	Local Government Engineering Department
MDB	Multilateral Development Bank
M&E	Monitoring and Evaluation
MFS	Mobile Financial Services
MIS	Management Information Systems
MoA	Ministry of Agriculture
MoDMR	Ministry of Disaster Management and Relief
MoF	Ministry of Finance
MoLGRD&C	Ministry of Local Government, Rural Development, and Cooperatives
MoWR	Ministry of Water Resources
NAP	National Adaptation Plan
NATP	Bangladesh National Agricultural Technology Program
NDC	Nationally Determined Contribution
NJLIP	Nuton Jibon Livelihood Improvement Project
NPV	Net Present Value
OHS	Occupational Health and Safety
O&M	Operations and Maintenance
PBA	Performance-Based Allocation
PIC	Project Implementation Committee
PIU	Project Implementation Unit
POM	Project Operational Manual
PPSD	Project Procurement Strategy for Development
PSC	Project Steering Committee
RCS	Regional Community Societies
RELI	Resilience, Entrepreneurship, and Livelihood Improvement Project
SAR	Search and Rescue
SEA/SH	Sexual Exploitation and Abuse / Sexual Harassment
SDF	Social Development Foundation
SNSP	Safety Net Systems for the Poorest
STEP	Systematic Tracking of Exchanges in Procurement
TEDs	Training, Exercises, and Drills
UAL	Universally Aligned
UN	United Nations
VDF	Village Development Funds
WASH	Water, Sanitation, and Hygiene



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**DATASHEET****BASIC INFORMATION**

Project Beneficiary(ies)	Operation Name		
Bangladesh	Bangladesh Sustainable Recovery, Emergency Preparedness and Response Project		
Operation ID	Financing Instrument	Environmental and Social Risk Classification	Process
P508058	Investment Project Financing (IPF)	Substantial	Urgent Need or Capacity Constraints(FCC)

Financing & Implementation Modalities

☐ Multiphase Programmatic Approach (MPA)	☒ Contingent Emergency Response Component (CERC)
☐ Series of Projects (SOP)	☐ Fragile State(s)
☐ Performance-Based Conditions (PBCs)	☐ Small State(s)
☐ Financial Intermediaries (FI)	☐ Fragile within a non-fragile Country
☐ Project-Based Guarantee	☐ Conflict
☐ Deferred Drawdown	☒ Responding to Natural or Man-made Disaster
☐ Alternative Procurement Arrangements (APA)	☐ Hands-on Expanded Implementation Support (HEIS)

Expected Approval Date	Expected Closing Date
14-May-2025	31-Dec-2030
Bank/IFC Collaboration	
No	

Proposed Development Objective(s)

The project development objectives are to (a) rehabilitate and enhance the resilience of critical public infrastructure and (b) support flood-affected households to strengthen livelihoods.

Components

Component Name	Cost (US\$)
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People Centric Resilient Infrastructure	173,000,000.00
Livelihoods Recovery and Enhancement Support	39,000,000.00
Community Recovery and Resilience to Disasters	19,000,000.00
Agricultural System Restoration	19,000,000.00
Project Administration, Management, Coordination, Monitoring and Evaluation	25,000,000.00
Contingent Emergency Response	0.00

Organizations

Borrower:	People's Republic of Bangladesh		
Contact	Title	Telephone No.	Email
Implementing Agency:	Local Government Engineering Department (Local Government Division, Ministry of Local Government, Rural Development and Co-operatives), Department of Agricultural Extension (Ministry of Agriculture), Department of Disaster Management (Ministry of Disaster Management and Relief), Bangladesh Water Development Board (Ministry of Water Resources), Social Development Foundation (Financial Institutions Division, Ministry of Finance)		
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PROJECT FINANCING DATA (US\$, Millions)

Maximizing Finance for Development

Is this an MFD-Enabling Project (MFD-EP)?	No
Is this project Private Capital Enabling (PCE)?	No

SUMMARY

Total Operation Cost	275.00
Total Financing	275.00



of which IBRD/IDA	270.00
Financing Gap	0.00

DETAILS

World Bank Group Financing

International Development Association (IDA)	270.00
IDA Credit	270.00

Non-World Bank Group Financing

Counterpart Funding	5.00
Borrower/Recipient	5.00

IDA Resources (US\$, Millions)

	Credit Amount	Grant Amount	SML Amount	Guarantee Amount	Total Amount
National Performance-Based Allocations (PBA)	120.00	0.00	0.00	0.00	120.00
Crisis Response Window (CRW)	150.00	0.00	0.00	0.00	150.00
Total	270.00	0.00	0.00	0.00	270.00

Expected Disbursements (US\$, Millions)

WB Fiscal Year	2025	2026	2027	2028	2029	2030	2031
Annual	0.50	20.00	45.00	50.00	59.50	60.00	35.00
Cumulative	0.50	20.50	65.50	115.50	175.00	235.00	270.00

PRACTICE AREA(S)



Practice Area (Lead)

Urban, Resilience and Land

Contributing Practice Areas

Social Protection & Jobs; Agriculture and Food; Transport; Education

CLIMATE

Climate Change and Disaster Screening

Yes, it has been screened and the results are discussed in the Operation Document

SYSTEMATIC OPERATIONS RISK- RATING TOOL (SORT)

Risk Category

Rating

1. Political and Governance	● Substantial
2. Macroeconomic	● Substantial
3. Sector Strategies and Policies	● Moderate
4. Technical Design of Project or Program	● Moderate
5. Institutional Capacity for Implementation and Sustainability	● Substantial
6. Fiduciary	● Substantial
7. Environment and Social	● Substantial
8. Stakeholders	● Moderate
9. Overall	● Substantial

POLICY COMPLIANCE

Policy

Does the project depart from the CPF in content or in other significant respects?

☐ Yes ☒ No

Does the project require any waivers of Bank policies?

☐ Yes ☒ No

ENVIRONMENTAL AND SOCIAL



Environmental and Social Standards Relevance Given its Context at the Time of Appraisal

E & S Standards	Relevance
ESS 1: Assessment and Management of Environmental and Social Risks and Impacts	Relevant
ESS 10: Stakeholder Engagement and Information Disclosure	Relevant
ESS 2: Labor and Working Conditions	Relevant
ESS 3: Resource Efficiency and Pollution Prevention and Management	Relevant
ESS 4: Community Health and Safety	Relevant
ESS 5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement	Relevant
ESS 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources	Not Currently Relevant
ESS 7: Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities	Relevant
ESS 8: Cultural Heritage	Not Currently Relevant
ESS 9: Financial Intermediaries	Not Currently Relevant

NOTE: For further information regarding the World Bank's due diligence assessment of the Project's potential environmental and social risks and impacts, please refer to the Project's Appraisal Environmental and Social Review Summary (ESRS).

LEGAL

Legal Covenants

Sections and Description

Section I.A.1, Schedule 2 of the FA: The Recipient shall and shall cause the Project Implementing Entity to establish within six (6) months of the Effective Date and maintain throughout the period of implementation of the Project, the Project Implementing Unit (PIU) at each of LGED, BWDB, DAE, DDM, and the Project Implementing Entity with the composition, mandate, functions, staffing and resources satisfactory to the Association.

Section I.A.2, Schedule 2 of the FA: The Recipient shall and shall cause the Project Implementing Entity to establish within six (6) months of the Effective Date and maintain throughout the period of implementation of the Project, the Project Steering Committee (PSC) at each of LGD, MoWR, MoA, MoDMR, and FID, with the composition, mandate, functions, staffing and resources satisfactory to the Association.

Section I.A.3, Schedule 2 of the FA: The Recipient shall and shall cause the Project Implementing Entity to establish within six (6) months of the Effective Date and maintain throughout the period of implementation of the Project, the Project Implementation Committee (PIC) at each of LGED, BWDB, DAE, DDM, and the Project Implementing Entity, with the composition, mandate, functions, staffing and resources satisfactory to the Association.

Section I.B.1, Schedule 2 of the FA: To facilitate the carrying out of Part 2 of the Project, the Recipient shall make the proceeds of the Credit available to the Project Implementing Entity under a subsidiary agreement between the Recipient, through FID, and the Project Implementing Entity, under terms and conditions satisfactory to the Association,



an amount not exceeding the equivalent of thirty-three million nine hundred thousand Special Drawing Rights (SDR 33,900,000) or such other amounts as shall be agreed from time to time (“Subsidiary Agreement”).

Section I.C.1, Schedule 2 of the FA: The Recipient, through each of LGED, BWDB, DAE, DDM, and the Project Implementing Entity, shall prepare and adopt a Project Operations Manual (POM) for the Project, in form and substance satisfactory to the Association within six (6) months of the Effective Date.

Section I.E.1, Schedule 2 of the FA: In order to ensure the proper implementation of contingent emergency response activities under Part 6 of the Project (“Contingent Emergency Response Part”), the Recipient shall and shall cause the Project Implementing Entity to ensure that: (a) a manual (“CERC Manual”) is prepared and adopted in form and substance acceptable to the Association, which shall set forth detailed implementation arrangements for the Contingent Emergency Response Part; (b) the Emergency Action Plan is prepared and adopted in form and substance acceptable to the Association; and (c) the Contingent Emergency Response Part is carried out in accordance with the CERC Manual and the Emergency Action Plan.

Conditions

Type	Citation	Description	Financing Source
Disbursement	Schedule 2, Section III.B	Notwithstanding the provisions of Part A of this Section, no withdrawal shall be made: (a) for payments made prior to the Signature Date, except that withdrawals up to an aggregate amount not to exceed seven hundred fifty thousand Special Drawing Rights (SDR,750,000) may be made for payments made prior to this date but on or after December 1, 2024, for Eligible Expenditures under Category (3), and upon completion of an environmental and social audit, in a manner satisfactory to the Association. (b) for Emergency Expenditures under Category (13), unless and until all of the following conditions have been met in respect of said expenditures: (i) (A) the	IBRD/IDA



		Recipient has determined that an Eligible Crisis or Emergency has occurred, and has furnished to the Association a request to withdraw Financing amounts under Category (13); and (B) the Association has agreed with such determination, accepted said request and notified the Recipient thereof; and (ii) the Recipient has caused the Project Implementing Entity to adopted the CERC Manual and Emergency Action Plan, in form and substance acceptable to the Association.	
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I. STRATEGIC CONTEXT

A. Project Strategic Context

- Bangladesh experienced extreme rainfall from mid-August to mid-September 2024, resulting in significant flooding that affected nearly six million people¹ and caused damages of at least US\$1.676 billion.²** The floods severely impacted 11 eastern districts.³ Infrastructure, housing, education, health, agriculture, fisheries, and livestock sectors were critically affected. Bangladesh had been facing a series of disasters, including Cyclone Remal (May 2024), the northeastern flood (June 2024), and the northern riverine monsoon flood (July 2024), which together impacted more than 18 million people.⁴ More broadly, the Global Climate Risk Index ranks Bangladesh as the seventh most affected country between 2000-2019⁵. In the past 70 years, around 20 percent of the country was flooded annually, while in recent years flood events affecting more than 30 percent of the country have become more frequent.⁶
- The Bangladesh Sustainable Recovery, Emergency Preparedness and Response Project (B-STRONG) aims to address recovery and reconstruction needs as well as strengthen disaster resilience of the country.** Flooding is a recurring phenomenon that affects approximately 1 million people annually in Bangladesh, severely undermining development gains. Therefore, the proposed project will follow a Build Back Better approach⁷ to reduce future flood risks. Further, the project is multi-sectoral, covering physical interventions and non-physical activities. Altogether, the project design ensures a holistic approach to recovery and resilience building, reducing future flood risks, supporting sustainable agriculture, economic development and preventing vulnerable populations from being left behind.

B. Sectoral and Institutional Context

- Frequent floods in Bangladesh – fluvial, pluvial, coastal, and flash floods – have had a significant impact on lives and livelihoods.** A deltaic country consisting of floodplains created by over 400 rivers and channels, 25 percent of the country is less than one meter above sea level, and 50 percent is less than 6 meters above sea level. Bangladesh is located at the foothills of the Himalayan Mountain range, which is the world's highest precipitation zone. During the monsoon season, winds from the sea heading towards land raise the water levels in the Bay of Bengal, inhibiting drainage from the rivers into the sea. As rainfall levels are also high during the monsoon seasons, river flows, local rain, and raised water levels for the Bay of Bengal result in flooding of vast areas. Large scale floods in 2017 killed 145 people and affected about 8 million people in 32 districts. Severe floods during the last week of June 2020 affected 5.4 million people.⁸ Recurring flooding in Bangladesh affects a greater share of the population than other natural hazards, with once every three to five years, up to two-thirds of the country is inundated.⁹
- The Government of Bangladesh (GoB) has shown its continued commitment to reducing climate and disaster risks.** The Bangladesh Delta Plan (BDP) 2100 provides a comprehensive strategy for sustainable management of water, ecology, environment and land resources in the context of their interaction with natural disasters and climate change. It delineates the country into seven hotspot areas¹⁰ to systematically address climate and disaster risks more effectively. Its implementation is guided by a 5-year plan, with the latest being the National Plan for Disaster

¹ Bangladesh Shelter Cluster. October 1, 2024. *Eastern Floods Shelter Damage and Needs Assessment*.

² World Bank. 2024. *Floods in eastern Bangladesh, August 2024: Global Rapid Post-Disaster Damage Estimation (GRADE) Report*.

³ Noakhali, Sylhet, Moulvibazar, Habiganj, Brahmanbaria, Cumilla, Feni, Lakshmipur, Khagrachhari, Chattogram, and Cox's Bazar.

⁴ UN OCHA. 2024. *Bangladesh: Cyclone and Monsoon Floods – May to August 2024*.

⁵ German watch. 2021. *Global Climate Risk Index 2021*.

⁶ FFWC. 2022. *Annual Flood Report 2021*. Processing & Flood Forecasting Circle, Flood Forecasting & Warning Centre, Bangladesh Water Development Board.

⁷ GFDRR. n.d. *Building Back Better in Post-Disaster Recovery Guidance Note*

⁸ International Federation of Red Cross and Red Crescent Societies. (May 5, 2021). *Final Report - Monsoon Floods: Bangladesh*.

⁹ World Bank Group. 2022. *Bangladesh Country Climate and Development Report*. CCDR Series; World Bank Group, Washington, DC.

¹⁰ Coastal zone, drought-prone areas, haor and flash flood areas, Chattogram hill tracts, river systems & estuaries, urban areas, and relatively less hazard prone areas.



Management 2021-2025. This plan emphasizes the importance of investing in disaster risk reduction to enhance resilience, following an “all-hazards” approach to address increasing climate risks.

5. **In August 2024, a massive flood occurred in the eastern part of the country.** The flood can be attributed to a very severe rainfall event between August 20-25, 2024. A peak daily rainfall of 200 to 300 millimeters was observed with a total rainfall of 600 to 800 millimeters within a week. This amount of rainfall is more than 2.5 times higher than average conditions. The floods had a quick onset ('flash floods') behavior due to the catchment topography. The upstream part of the catchment where Feni, Noakhali and Cumilla are located is hilly and its altitude is much higher. The catchments are small and short, resulting in quick runoff of rainfall in downstream direction. The floods affected nearly six million people resulting in 74 fatalities, stranding 1.26 million families, and forcing over 500,000 people to take refuge in disaster shelters.
6. **Other significant natural hazards happened in different parts of the country the preceding months:** the Northern Riverine Monsoon Flood in July (affecting 5.1 million people), the Northeastern Flood in June (affecting 3.7 million people), and Cyclone Remal in the southwest in May (affecting 4.6 million people). Cyclone Remal was the second most impactful disaster in 2024 in terms of damages, with total losses estimated to be more than US\$600 million.¹¹
7. **The total direct physical asset damages from the August 2024 Flash Flood are estimated at US\$1.676 billion.**¹² They include infrastructure (US\$640 million), agriculture (US\$468 million), residential (US\$403 million), and non-residential buildings (US\$165 million). Highest damages are observed in Noakhali (US\$508.8 million), Cumilla (US\$362.6 million), and Feni (US\$312.7 million). Infrastructure damage includes 3,700 kilometers of fully damaged and 14,000 kilometers of partially damaged roads, 87 kilometers of damaged embankments, 17 sections of breached flood control embankments, 28,000 fully damaged and 319,000 partially damaged houses, 423 fully damaged and 1,097 partially damaged bridges and culverts, as well as 430 damaged healthcare facilities. Around 3,000 education facilities were impacted, affecting 1.4 million children, including 749,390 girls in Chattogram Division alone.¹³
8. **Food security and livelihoods remain fragile.** According to the Department of Agricultural Extension (DAE), over 1.3 tons of standing crops were lost, and around 0.6 million hectares of land was affected. The flood caused a production loss of over half a million tons of rice according to the Food Planning and Monitoring Unit (FPMU),¹⁴ impacting two million farmers. Fisheries and livestock losses were reported at US\$135 million and US\$34 million, respectively. Flooding of farmland is expected to have a lasting impact on food security. Up to 40 percent of the affected population resorted to negative coping mechanisms and 85 percent of affected households anticipate losing their secure income.¹⁵ Over 85 percent of rural households depend on agriculture for their livelihoods, and with increasing reliance on groundwater and pesticides to mitigate declining agricultural outputs, food and nutritional security is at risk.¹⁶
9. **The scale and magnitude of the flood damages and continuing vulnerability of the country to climate-related disasters require significant investments to rehabilitate and strengthen basic services and infrastructure.** Although relief efforts have been put forward by the GoB, United Nations (UN) agencies, and Development Partners (DPs), the funding is insufficient to cover the flood impacts. The Humanitarian Coordination Task Team, co-chaired by the

¹¹ Needs Assessment Working Group Bangladesh. 2025. *Snapshot of Disasters in Bangladesh 2024 - D-Form Analysis*.

¹² World Bank. 2024. *Floods in eastern Bangladesh, August 2024: Global Rapid Post-Disaster Damage Estimation (GRADE) Report*.

¹³ Inter-cluster Coordination Group (ICCG). August 30, 2024. *Bangladesh: Eastern Flash Floods - Situation Report No. 02*.

¹⁴ FPMU. *Bangladesh Food Situation Report April-June 2024*. Dhaka.

¹⁵ Food and Agriculture Organization. September, 2024. *Bangladesh: Impact of the floods on agricultural livelihoods and food security in the eastern part of the country – DIEM impact report*. Rome.

¹⁶ World Bank. 2016. “Bangladesh: Growing economy through advances in agriculture.” *World Bank Results*.



Secretary of the Ministry of Disaster Management and Relief (MoDMR) and the UN Resident Coordinator, updated its Multi-hazard Humanitarian Response Plan (HRP) on September 29, 2024, to include the 2024 Eastern floods. The funding requested for the August 2024 Eastern Flash Floods alone is US\$71 million.

- 10. In addition to financing gaps for immediate flood response, there is an even larger gap for medium- and long-term recovery and reconstruction.** Reconstruction of damaged assets would require more than US\$1.6 billion and could take years. Damaged public infrastructure, such as roads and electricity, hampers livelihoods and economic activities. Unrepaired embankments and drainages drastically reduce protection from disasters, implying that higher impacts may occur even with less severe disasters. These gaps need to be urgently addressed, in addition to supporting the affected population to get back on their feet. Research posits the importance of financial capital, assets, and networks as foundations for resilience to disasters.^{17,18} Support to enhanced and diversified livelihoods, including linkages to producer networks and markets, will enable households to recover, revitalize the local economy, and contribute to building longer-term community resilience. The proposed project is multi-sectoral in order to provide holistic recovery and will support sectors that are not sufficiently covered by existing initiatives.
- 11. By following a Build Back Better approach to medium- and long-term recovery and reconstruction, the government can build institutional and technical capacity for resilient infrastructure investments.** The rehabilitation of damaged infrastructure presents an opportunity to improve the quality of the infrastructure in the affected areas, as well as promote disaster resilience across the country. For example, rural roads can be upgraded to better cope with future climate extremes, while embankments and flood control structures can be rehabilitated to higher design standards.
- 12. The proposed B-STRONG project will contribute to achieving the objectives of the BDP 2100, which highlights the need to reduce vulnerability to floods, and build resilience to climate change and other delta challenges.** The first of the six goals of the BDP 2100 is to ensure safety from floods and climate change-related disasters. The flood risk management strategies of the BDP 2100 aim to mitigate the most undesired effects of large and extreme floods and enable flood-affected people to recover quickly in the aftermath of flood events. These strategies include a principle of climate resilient development through participatory processes and focus on safeguarding vulnerable communities through flood and storm surge proofing of critical infrastructure. However, this flood event serves as another example of how inadequate infrastructure can lead to significant impacts from floods and other similar disasters. Investing in sustainable solutions by implementing BDP 2100 is an urgent and critical need.
- 13. The project aligns with the World Bank Country Partnership Framework (CPF) FY23-27 (Report No. 181003-BD), discussed by the World Bank Board of Executive Directors on April 27, 2023.** It will support the CPF's High-Level Outcome B (Improved Socioeconomic Inclusion) and C (Enhanced Climate and Environmental Resilience). People benefitting from enhanced resilience to riverbank erosion or inundation damage, length of road network in good and fair conditions, and beneficiaries of economic inclusion program, are among the CPF's progress indicators that will be supported by this project.

II. PROJECT DESCRIPTION

A. Project Development Objective

- 14.** The project development objectives are to (a) rehabilitate and enhance the resilience of critical public infrastructure and (b) support flood-affected households to strengthen livelihoods.

¹⁷ Azad, M. & Pritchard, B.. 2022. "Financial capital as a shaper of households' adaptive capabilities to flood risk in northern Bangladesh." *Ecological Economics* 195.

¹⁸ Islam R., Walkerden, G. 2014. "How bonding and bridging networks contribute to disaster resilience and recovery on the Bangladeshi coast." *International Journal of Disaster Risk Reduction* 10 (Part A).



B. Theory of Change and PDO Indicators

15. Progress towards the PDO will be measured through the following indicators:

- People benefitting from climate resilient infrastructure (Number of people).¹⁹
- People benefitting from climate resilient planning, preparation, surveillance, and/or response (Number of people).²⁰
- Beneficiaries of cash-based interventions (Number of people).²¹
- Beneficiaries of economic inclusion programs (Number of people).²²
- People benefitting from access to improved agriculture production system (Number of people).²³

16. The project's theory of change and result chain are presented in Figure 1.

C. Project Beneficiaries

17. The project will benefit over 1.6 million people across selected flood-affected districts in the Chattogram and Sylhet Divisions (Noakhali, Sylhet, Moulvibazar, Habiganj, Brahmanbaria, Cumilla, Feni, Lakshmipur, Chattogram, and Cox's Bazar). Around 175,000 people will be protected by flood protection infrastructure, after the structures were damaged by the Eastern Flash Floods. Approximately 237,000 will benefit from increased access to multipurpose disaster shelters. Approximately 88,000 direct beneficiaries in 550 villages in selected districts will strengthen their livelihoods, benefitting 374,880 people. Approximately 8,857 people from four districts will directly benefit from temporary livelihood opportunities, benefitting 37,731 people.²⁴ Production and productivity of at least 65,000 farm households in 430 unions will be enhanced through the adoption of projects promoting production technologies and access to improved irrigation and storage facilities.

D. Project Components

18. The proposed project comprises six components. Table 1 shows the breakdown of the budget per component. The project financing includes both the country's Performance-Based Allocation and the Crisis Response Window (CRW). The CRW financing is sought as a last resort financing to support Bangladesh's flood response and recovery, complementing the Bank's overall crisis response.

Table 1 Breakdown of costs per component (US\$ million)

Component	Total
Component 1: People Centric Resilient Infrastructure	173
Component 2: Livelihoods Recovery and Enhancement Support	39
Component 3: Community Recovery and Resilience to Disasters	19
Component 4: Agricultural System Restoration	19
Component 5: Project Administration, Management, Coordination, Monitoring and Evaluation	25 ²⁵
Component 6: Contingent Emergency Response	0
<i>Total</i>	275

¹⁹ This is a Corporate Result Indicator (CRI). In this project, it includes number of people with access to at least one climate resilient infrastructure as a result of this project: emergency evacuation rural roads rehabilitated, emergency evacuation bridge rehabilitated and reconstructed.

²⁰ This is a CRI. In this project, it includes people benefitting from flood risk reduction through rehabilitation, repair and construction of flood protection structures and reduced exposure to climate-related hazards due to the access to multipurpose disaster shelters.

²¹ This is a CRI. Activities under Component 3 will contribute to this indicator.

²² This is a CRI. Activities under Component 2 will contribute to this indicator.

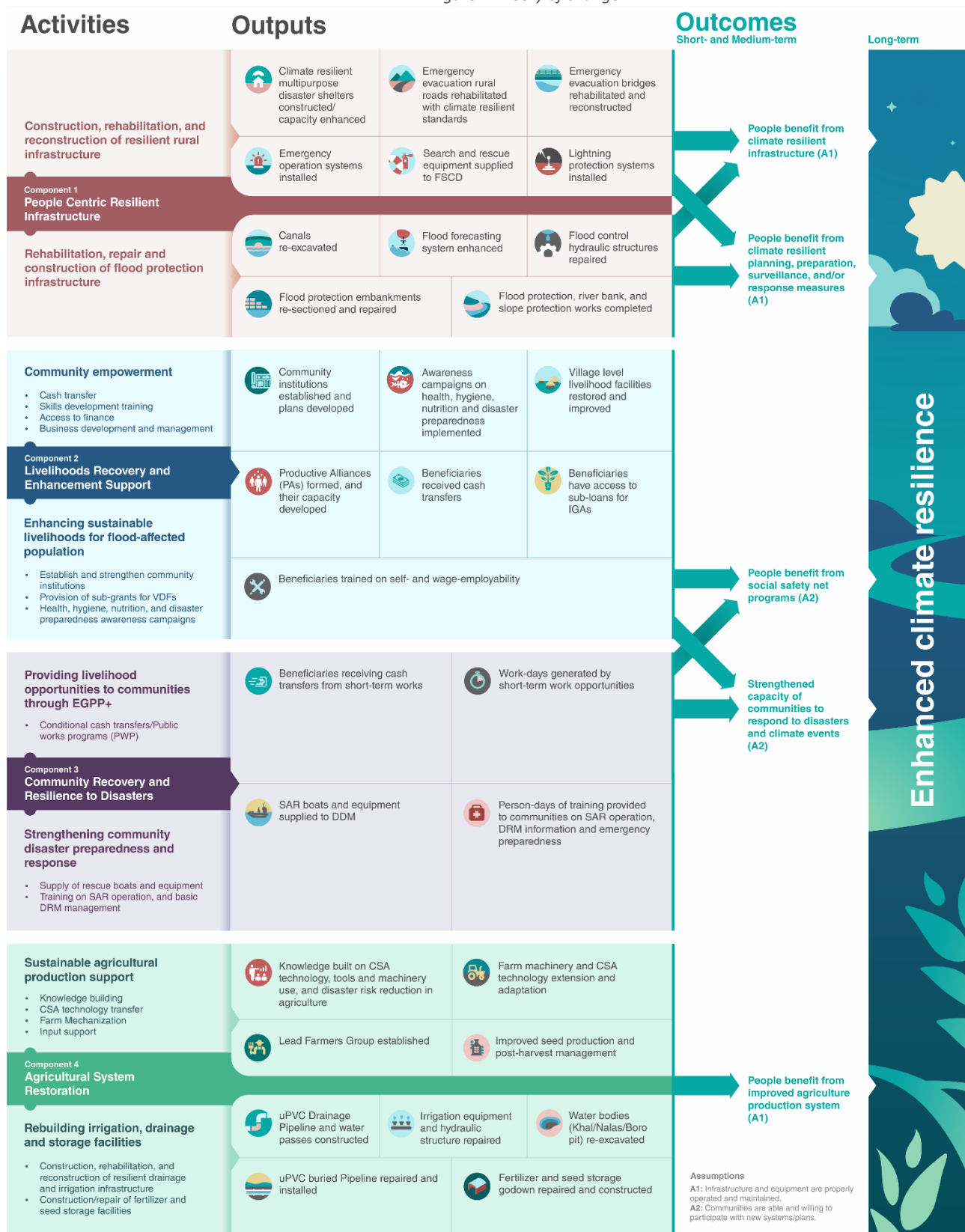
²³ This is a custom CRI. This covers people benefitting from restoration of agricultural production, access to technology and extension services, rehabilitation of irrigation and drainage networks, seed and fertilizer storage facilities for better agri-system management.

²⁴ The average household size is 4.26 person/household in Bangladesh. Directly offering livelihoods support to one family member will benefit an entire family.

²⁵ This includes US\$5 million financing from the GoB.



Figure 1 Theory of change





19. Components 1 and 4 will follow a framework approach, where the exact number and location of investments will be finalized in consultation with the GoB and relevant stakeholders during implementation, owing to the dynamic environment with evolving needs, stakeholders, and programs. The framework approach will follow two principles. First, activities will be identified through a multi-criteria analysis.²⁶ Second, all activities will be screened in compliance with Environmental and Social Standards (ESS) and activities with high Environmental and Social (ES) risk will be excluded, as articulated in the Environmental and Social Management Framework (ESMF).

Component 1: People Centric Resilient Infrastructure (US\$173 million equivalent)

Subcomponent 1.1: Resilient Rural Infrastructure (US\$127.5 million equivalent)

20. **Implemented by the Local Government Engineering Department (LGED), this subcomponent will finance the construction, rehabilitation, and reconstruction of flood-affected rural infrastructure and facilities under LGED's jurisdiction**, in selected districts, including: (i) construction of climate resilient multipurpose disaster shelters;²⁷ (ii) capacity enhancement and rehabilitation of existing disaster shelters by extending one floor with climate proofing connecting roads (above flood level);²⁸ (iii) improvement of damaged existing schools affected by flood to climate resilient standards;²⁹ (iv) rehabilitation of rural roads with minor enhancement of capacity to climate-resilient standards;³⁰ (v) rehabilitation and reconstruction of existing bridges and culverts with minor capacity expansion;³¹ (vi) slope protection on road alignment to protect against extreme precipitation and wave action exacerbated by climate change; (vii) supply and installation of lightning protection systems to decrease vulnerability to lightning strikes resulting from extreme hydro-metrological events; (viii) supply of search and rescue equipment for the Fire Service and Civil Defence (FSCD) for providing emergency response³² during extreme climate events (floods, landslides, cyclones, fires); and (ix) supply and installation of Emergency Operation Systems³³ for *upazilas*.

21. **Build Back Better principles inform the subcomponent's design to ensure development and resilience benefits.** The disaster shelter design will follow the Bangladesh National Building Code (BNBC), which include universal accessibility; Water, Sanitation, and Hygiene (WASH) facility incorporation; and gender sensitive and climate resilient design, such as fire safety standards and energy efficiency compliance.³⁴ For roads protective infrastructure will be included. This will make the roads more all-weather resistant and reduce flood risks along the road network. The rehabilitation and reconstruction of bridges will also improve access for rural communities and facilitate evacuation during emergencies, while taking into account the presence of existing canals, streams, and drainages, as to not

²⁶ The analysis relies on the following key criteria: flood area, potential beneficiaries (the presence of agriculture, aquaculture, and densely populated area), urgency (whether the damage process has worsened since the flood event), availability of construction materials, high-level cost estimate, stakeholder priorities (urgency of requests from the respective agencies), land ownership, availability of contractors, and availability of additional funding

²⁷ The primary purpose of disaster shelters is to offer a safe haven for communities and livestock from floods, cyclones, storm surges, and strong winds, while their use as schools for the remainder of the year serves a dual purpose of ensuring maintenance and fulfilling educational needs, with these activities being secondary to the disaster shelters' main function. The construction and capacity expansion of multipurpose disaster shelters includes rainwater harvesting, and renewable energy backup systems with solar power as well as energy efficient lights and fans to ensure shelter functionality during floods, and cyclones.

²⁸ Refer to footnote 26 with regards to purpose of capacity expansion of disaster shelters and other details.

²⁹ To withstand the impact of high wind, flooding, extreme precipitation, and extreme temperature.

³⁰ To increase readiness and resilience to sea level rise, extreme precipitation, flooding, storm surges and landslides, including associated roadside drainage network to reduce the risk of flooding and landslides. The overall resiliency of road infrastructure will be enhanced by sealing the surface of roads, constructing roadside drainage networks. These activities will help prevent washouts while maintaining critical access for logistics and resources in rural areas, particularly in facilitating movement and access to essential services (connecting farmers to markets and rural residents to disaster shelters, schools, hospitals, or social services) during all weather conditions and emergencies.

³¹ Aligned with the roadside drainage network to drain the increased surface run-off from extreme precipitation and flooding.

³² Strengthening search & rescue operations through ensuring adequate emergency response equipment, and training the first responders and the FSCD in the onset of extreme hydrometeorological events and fire.

³³ For strategic coordination and decision-making capacities for evacuation and emergency preparedness during and onset of extreme climate events (floods, cyclones, landslides and lightning events) and other disasters. This would also support strengthening community based early-warning systems for hydrometeorological hazards.

³⁴ BNBC 2020 focused on energy efficiency through daylight use and the inclusion of renewable energy in line with energy efficiency and conservation master plan.



reduce their design capacity. This subcomponent will also support the planning for critical operations and maintenance (O&M) needs of the infrastructure beyond the project period.

Subcomponent 1.2: Flood Protection Rehabilitation (US\$45.5 million equivalent)

22. Implemented by the Bangladesh Water Development Board (BWDB), this subcomponent will support the rehabilitation, repair, and construction of flood protection infrastructure through a combination of interventions in selected districts³⁵ including (i) flood protection embankment re-sectioning and repair works; (ii) canal re-excavation works³⁶ to increase the stormwater drainage capacity during flood; (iii) flood control hydraulic structure reparation works; (iv) flood protection river bank and slope protection works; and (v) improvement of the flood forecasting system through expansion of the flood models, forecasting facilities, and technical capacity building. This subcomponent will be implemented across selected flood affected districts where flood protection infrastructure, and control structures were significantly damaged. The repair and rehabilitation works are critical to protect communities and livelihoods from future flooding. Canal re-excavation works will reduce flood risks that are amplified due to significant sedimentation in the canals and will increase stormwater drainage capacity. Flood control hydraulic structure-related works include the repair of damaged inlet and outlet structures, sluice gates, water regulators, and drainage. The collapse of such control structures has changed the tidal regime and increased flood risks.

23. To maximize impact, the project will focus on the most vulnerable and flood-impacted hotspots, ensuring that resources are allocated effectively. Priority will be given to the flood protection structures that are essential for mitigating severe and prolonged waterlogging, as well as minimizing the recurrence of overtopping, breaches, and stagnant floodwaters that disrupt livelihoods and access to services. The interventions will be spread throughout the affected districts, but careful consideration will be given to the relative vulnerability and needs of each district to ensure equitable resource distribution. Moreover, the project will enhance and apply the numerical tools for monitoring and morphological assessment developed under the Coastal Embankment Improvement Project Phase 1 (CEIP-I) (P128276) for practical applications in climate-resilient investments. This subcomponent will also support the planning for critical O&M needs of the flood protection infrastructure beyond the project period.

Component 2: Livelihoods Recovery and Enhancement Support (US\$39 million equivalent)

24. This component focuses on restoring and enhancing livelihoods of the flood affected population through services that support economic opportunities for affected households. Social Development Foundation (SDF) will implement this component, drawing on their extensive community driven development (CDD) experience,³⁷ to support approximately 88,000 beneficiary households across selected districts.³⁸ SDF will facilitate the establishment and strengthening of community institutions³⁹ at the village, cluster, and regional levels, which will play a central role in strengthening resilience of vulnerable groups to natural and man-made shocks through enhanced and diversified livelihoods. This component works synergistically with the infrastructure interventions component. While Component 1 restores resilient basic and flood protection infrastructure to reduce physical vulnerabilities, this component empowers flood-affected households with enhanced livelihood opportunities and institutional support. Together, these components form a holistic approach to recovery and resilience building, supporting both the physical and socio-economic aspects of disaster resilience.

³⁵ Brahmanbaria, Cumilla, Feni, Noakhali, and Lakshmipur.

³⁶ Canals under BWDB jurisdiction.

³⁷ CDD is a modality of project design and delivery which transfers decision making power and financial and technical resources directly to communities. A creation of local group or committee to manage the project, external facilitation support, and contribution from community in cash or labor are common features.

³⁸ Noakhali, Sylhet, Moulvibazar, Habiganj, Brahmanbaria, Cumilla, Feni, Lakshmipur, Chattogram, and Cox's Bazar.

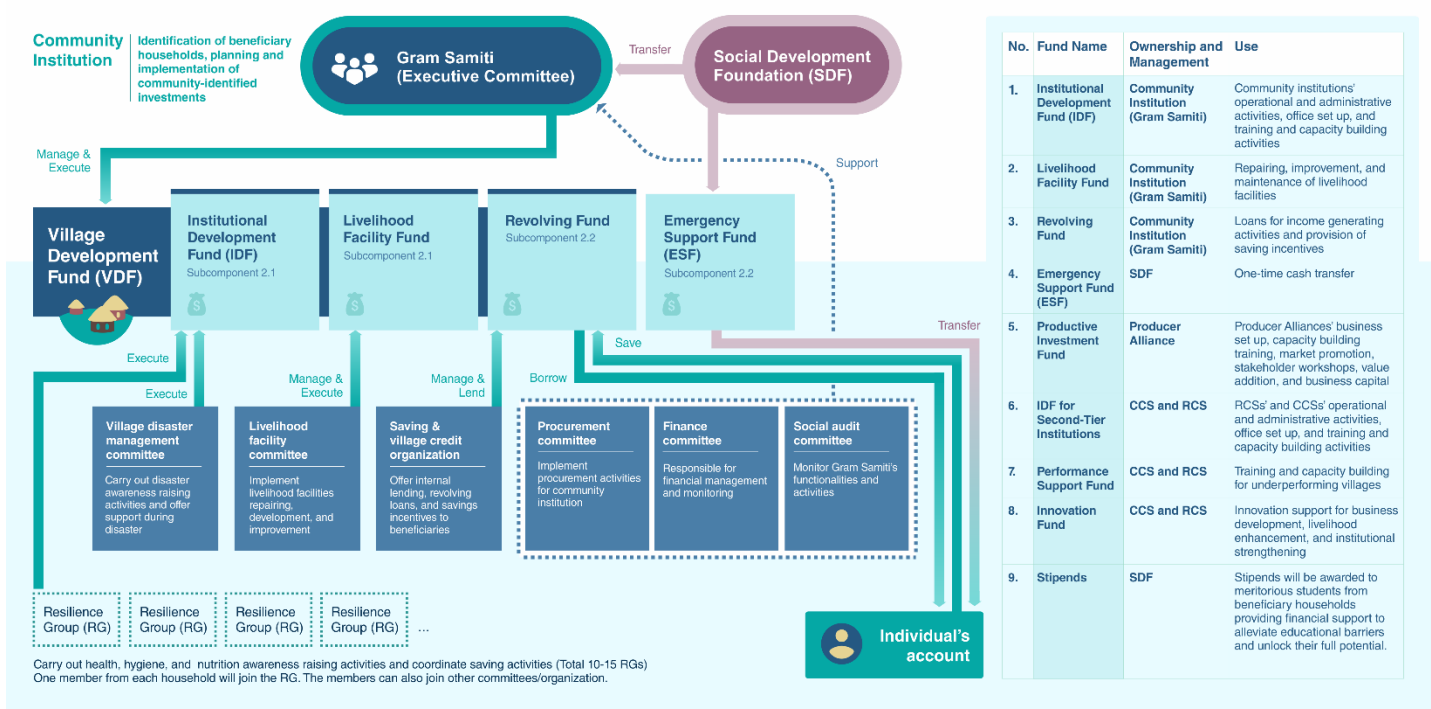
³⁹ Community institutions are newly created organization for the purpose of implementing Component 2 and are independent from existing formal or informal governing entities. A collaborative relationship with local government entities will be explored during implementation.



Subcomponent 2.1: Community Empowerment and Institutional Strengthening (US\$16.2 million equivalent)

25. This subcomponent will support a program of activities aimed at establishing and strengthening community institutions, including: (i) mobilization efforts at the community level to form community institutions, including Cluster Community Societies (CCS) and Regional Community Societies (RCS); (ii) provision of sub-grants to the Village Development Fund (VDF) and the CCS and RCS to implement community plans, institutional development activities, and activities to improve climate-resilient livelihood facilities; and (iii) awareness-raising activities related to hygiene, nutrition, and disaster preparedness. Community institutions will have 90 percent female representation, with the positions of president, secretary, and cashier reserved for women. Women leaders and other members will receive training and capacity building to foster their leadership and fulfill their roles. Project participation agreements signed between community institutions and SDF will detail roles and responsibilities, including women's quota in leadership positions. Community institutions will prepare community plans, based on community-identified needs and priorities, that will be financed from the VDF, including activities under subcomponent 2.2. The development, operation, and implementation of community institutions and community plans will follow guidelines set out in the Community Operations Manual (COM). Community institutions' set up and VDF sub-funds (Figure 2) and funds to be developed, including those outside of the VDF, are provided below.

Figure 2 Community institutions diagram



26. Community institutions will be clustered at the sub-district level (as CCS) and the regional level (as RCS) to facilitate collaboration; support multi-level institutional growth and governance; and aggregate service demand. To ensure sustainability of community institutions beyond project duration, RCS and CCS will be formally registered as legal entities and will be supported through membership fees. SDF will facilitate their relationship with financial institutions for other financing needs. Sustainability efforts will be made through consultation and coordination with local government bodies to ensure continuity of support.

Subcomponent 2.2: Enhancing Sustainable Livelihoods (US\$22.8 million equivalent)



27. This subcomponent will offer a package of services to enhance earning opportunities of beneficiaries to build their resilience to shocks, including (i) provision of one-time cash transfers, sub-loans (financed from the VDF), and stipends; (ii) skills development training on market-driven needs and financial literacy; and (iii) provision of sub-grants for the Productive Investment Fund to producer alliances to support business development, market linkages, and product promotion. Beneficiaries will be flood-affected and vulnerable groups, including poor and extreme poor, unemployed and underemployed youth, and those who need special care, support, protection due to their age, disability, or risk of mistreatment or neglect. They will be identified through a community participatory process, following eligibility guidelines laid out in the COM and based on multifaceted dimensions of poverty. Services will be tailored to the specific needs of beneficiaries and include one-time cash transfers to help meet basic needs, access to sub-loans for income generating activities (IGAs), and stipends for meritorious students. Skills development and training will support households to engage in improved and diversified livelihoods, which will help avoid negative coping mechanisms during shocks. All transactions will be through formal banking and financial services channels, with efforts to ensure they are made to the woman's financial account. Activities will be conducted in close consultation with community members, relevant line ministries, and extension services, including the DAE, Department of Livestock, and Department of Fisheries.

Component 3: Community Recovery and Resilience to Disasters (US\$19 million equivalent)

28. This component will be implemented by the Department of Disaster Management (DDM) in the selected districts.⁴⁰

Subcomponent 3.1: Providing Livelihood Opportunities to Communities through EGPP+ (US\$10.5 million equivalent)

29. This subcomponent will support the Employment Generation Program for the Poorest Plus (EGPP+), which is the shock-responsive window of DDM's national poverty-targeted public works program. The program has been operating in Cox's Bazar district since FY22 and five other districts since FY23. The subcomponent will provide temporary employment in three flood affected districts (Chattogram, Feni and Brahmanbaria), following the EGPP+ Implementation Guidelines.⁴¹ Temporary employment activities will include public works (e.g., rural road repair, cleaning, reforestation) and community services (e.g., information campaigns, training, and social mobilization to address GBV). The wages and stipends paid through the program will provide critical income support and employment to those most impacted by the floods. All EGPP+ data (i.e., beneficiary profiles, work attendance, payments) will be administered under the EGPP+ Management Information System (MIS).

Subcomponent 3.2: Strengthening Community Disaster Preparedness and Response (US\$8.5 million equivalent)

30. This subcomponent will support community preparedness and response to disasters through (i) the supply of rescue boats and equipment, and (ii) training, exercises, and drills (TEDs). The rescue boats will enhance the operational efficiency of flood response activities, by enabling faster evacuation of at-risk populations; reducing casualties; and facilitating the rapid delivery of food, clean water, and medical aid to stranded communities. The TEDs program will build critical capacities in disaster risk management of communities, volunteers, and local authorities in disaster-prone areas. The training activities will include, but are not limited to, emergency operation planning, emergency operation center management, and climate-resilient infrastructure. The program will train approximately 2,400 volunteers and Disaster Management Committee members in 37 flood prone districts on Search & Rescue (SAR) operations and basic disaster risk management. The participants will receive practical training in boat operation for emergency response, contributing to broader local and national SAR efforts.

Component 4: Agricultural System Restoration (US\$19 million equivalent)

⁴⁰ Feni, Brahmanbaria, Cumilla, and Chattogram.

⁴¹ The EGPP+ Implementation Guidelines were issued by MoDMR on January 31, 2021, to govern the implementation of the program window as part of the Safety Net Systems for the Poorest (SNSP) Project. The World Bank provided no objection on February 2, 2024.



31. This Component will improve agricultural livelihoods to enhance food security and nutrition. DAE, together with the Bangladesh Agricultural Development Corporation (BADC) as grantee, will implement the component in 40 upazilas of selected flood affected districts⁴² to support at least 65,000 farm households. The proposed interventions are aligned with government policies and strategies, address critical needs in agriculture,⁴³ and build long-term resilience against future climate-related challenges.

Subcomponent 4.1: Sustainable Agricultural Production Support (US\$9.3 million equivalent)

32. Implemented by the DAE, this subcomponent aims to enhance production through technology adaptation and foster agricultural mechanization. The support will offer to the farming communities: (i) inputs support, promote high yielding, climate resilient, water saving and sustainable crop production technologies, management methods and practices through technology demonstrations and arranging field days; (ii) technology training to farmers to enhance performance and capacity building of farmers, and entrepreneur; (iii) agricultural machinery support with the objectives to reduce laborious work, compensate labor shortage, enhance productivity and to promote farm mechanization;⁴⁴ (iv) establishment of polyshed house and seed village to supply quality seeds and seedlings; (v) establishment of community vermicompost and tricho-compost plants to improve soil health management; (vi) homestead gardening to support women, community gardening targeting the vulnerable and bringing fallow land under crop production; and (vii) support for the establishment of farmer groups to disseminate sustainable agricultural crop production technologies.

Subcomponent 4.2: Rebuilding Irrigation, Drainage and Storage Facilities (US\$9.7 million equivalent)

33. This subcomponent will be implemented by the DAE as part of a strategic alliance, with the BADC as grantee. A Memorandum of Understanding (MoU) in the form of Grant Agreement between the DAE and BADC will provide Grant Transfer from DAE to BADC, enabling BADC to carry out the activities. This subcomponent will rehabilitate and rebuild the damaged irrigation infrastructure and reconstruct drainage networks and storage facilities under BADC's jurisdiction. Activities include: (i) repair and reconstruction of damaged irrigation equipment, and infrastructure; (ii) re-excavation of filled water bodies, construction of underground drainage channels, and water passes to increase surface water availability for dry season supplementary irrigation; (iii) construction of uPVC drainage pipe line and water passes to increase drainage capacity during flood or excessive rainwater; (iv) construction of uPVC pipe line for buried pipes in existing scheme to increase irrigation water use efficiency, reduce water loss and maintenance work⁴⁵; and (v) construction and renovation of damaged fertilizer and seed storage godowns, construction of Sunning Floor, and Loading-Unloading Shades with climate-resilient standards to withstand the impact of high wind, flooding, extreme precipitation, and extreme temperature. These reconstruction works will secure production of food grains in the selected districts.

34. The infrastructure, drainage, and irrigation⁴⁶ works of BADC are largely confined within the agency's own mandates, including small-scale minor irrigation infrastructure support with pumping options, and preservation of quality seeds and fertilizer for farmers. Considering the limited resources and benefits to the affected communities,

⁴² Brahmanbaria, Cumilla, Feni, Noakhali, and Lakshmipur.

⁴³ Based on the FAO impact assessment study (September 2024), a large portion of affected farmers expressed the need for seeds, fertilizer and other crop inputs, including machinery, irrigation, fuel, transport, labour, pesticides and tools.

⁴⁴ Machinery distribution will follow AIF2 or AIF3 approach of World Bank supported closed National Agricultural Technology Program (NATP) II project which will be detailed out in POM.

⁴⁵ The rehabilitation/expansion of the irrigation system includes additional features and design considerations to ensure the system's resilience to the physical risks from climate change impacts with extreme precipitation patterns and flooding.

⁴⁶ BADC, BWDB, and LGED, are all involved in different sized irrigation schemes. BADC is mandated for small-scale minor irrigation (largely covering 100-200 hectares per scheme) with pumping, infrastructure and technology support while BWDB handles large sized schemes with the capacity of over 1000-hectare coverage and delivers water to tertiary head gates. The LGED irrigation schemes are small and medium sized, cover below 1000 hectares with no pumping options or infrastructure support, and normally create water reservoirs to facilitate a group of specific farmers.



the component will cover 40 upazilas of the selected districts. This subcomponent will also support the planning for critical O&M needs of the infrastructure under this subcomponent beyond the project period.

Component 5: Project Administration, Management, Coordination, Monitoring and Evaluation (Total financing: US\$25 million equivalent; IDA financing US\$20 million equivalent; GoB financing US\$5 million equivalent)

Subcomponent 5.1: Project Management, Coordination, Monitoring and Evaluation of LGED, BWDB, and DAE (IDA financing US\$ 13 million equivalent)

Subcomponent 5.2: Project Management, Monitoring and Evaluation of SDF (IDA financing US\$6 million equivalent)

Subcomponent 5.3: Project Management, Monitoring and Evaluation of DDM (IDA financing US\$1 million equivalent)

35. This component will support the day-to-day project administration, management, coordination, monitoring, and evaluation of activities by the Project Implementation Units (PIUs). Financing will be provided for: (i) salaries for contractual staff and consultant fees (for all), particularly SDF for carrying out supervision and hands-on support to the newly formed community institutions; (ii) technical assistance and capacity building activities (for LGED, BWDB, and DAE); (iii) goods and equipment (for all); (iv) incremental operating costs (for all); and (v) firm consultancy services for design and supervision and monitoring and evaluation (for LGED, BWDB, and DAE) and EGPP+ support, supervision, and monitoring and evaluation (for DDM).

Component 6: Contingent Emergency Response (US\$0 million)

38. This component will ensure provision of immediate response to an eligible crisis and contribute to immediate and effective response. Any unused balance under the other components can be reallocated to this Contingent Emergency Response Component, in the event of an emergency.

E. Role of Partners

39. The project has been designed in a coordinated manner with UN agencies and development partners.

Name of Partner	Nature of Involvement /Description
UN agencies, DPs, international and domestic NGOs	The Humanitarian Coordination Task Team (co-chaired by the Secretary of MoDMR and the UN Resident Coordinator) launched a Humanitarian Appeal and the Multi-hazard HRP for the Eastern Flash Floods. The funding request totaled US\$71 million. The total external financing mobilized in the initial humanitarian response as of October 2024 was around US\$12.7 million, focusing on cash and food assistance, WASH, and medical supplies. The proposed project will coordinate to minimize duplication and maximize synergies in flood recovery support.

F. Lessons Learned and Reflected in the Project Design

40. The project design reflects the key findings of the Independent Evaluation Group's report on Reducing Disaster Risks from Natural Hazards.⁴⁷ Achieving resilient infrastructure requires a comprehensive approach including robust O&M plans, meaningful community engagement, and a focus on addressing systemic risks.

41. Addressing O&M in resilient infrastructure projects is essential for long-term resilience: Failure to adequately address O&M deficiencies is especially critical for Bangladesh, where many basic infrastructures are often exposed to flooding and flood protection structures are at risk of erosion. Unless budgets and arrangements for O&M are adequate, a cycle of build-neglect-repair/replace is likely to continue, which undermines resilience.

⁴⁷ IEG. August 24, 2022. *An Evaluation of the World Bank's Support - Fiscal Years 2010–20*.



- 42. Community engagement in the development of resilient infrastructure systems is associated with effective construction or rehabilitation and likelihood of sustainability.** Actively engaging communities in consultations on local infrastructure systems, delegating decision-making responsibility to community organizations to foster ownership and ensure the infrastructure meets local needs and is more likely to be maintained.
- 43. A realistic objective and a simple, flexible design are critical for emergency operations.** Emergency operations have higher chances of success if they remain focused and avoid over-ambitious development objectives and targets, as evidenced in Jamaica Hurricane Dean (P109575), and Madagascar Emergency Support to Critical Education, Health and Nutrition (P131945) emergency response projects.
- 44. A holistic approach to resilience that considers the interconnected infrastructure elements can effectively address system-wide vulnerabilities and enhance overall durability.** The success of the Uttarakhand Disaster Recovery project (P146653) was partly attributed to interconnected infrastructure investments, where each element was expected to contribute to overall resilience (e.g., resilient materials coupled with adequate drainage systems prevented waterlogging and damage to roads).
- 45. Developing and strengthening community institutions is key for empowerment of rural women.** The Nuton Jibon Livelihood Improvement Project (NJLIP) (P149605) mobilized a high share of women beneficiaries into community groups. Since men usually work outside while women stay at home for household chores, eventually men encouraged women to participate in community-level project activities. Poor and extreme poor community members were involved in planning, managing and implementing village-level activities and decision making.

III. PROJECT IMPLEMENTATION

A. Institutional and Implementation Arrangements

- 46. The GoB will have overall responsibility for project implementation and management through its LGD, Ministry of Local Government, Rural Development and Co-operatives (MoLGRD&C); Ministry of Water Resources (MoWR); Ministry of Agriculture (MoA); MoDMR; and Financial Institutions Division (FID), Ministry of Finance (MoF).** The project will be implemented by LGED, BWDB, DAE, DDM, and SDF through five PIUs. The corresponding implementing agencies for component 1 are LGED for subcomponent 1.1, and BWDB for subcomponent 1.2. SDF, an autonomous organization under the FID, MoF, will be the implementing agency of component 2. Component 3 will be implemented by DDM. Subcomponent 4.1 will be implemented by DAE. Subcomponent 4.2 will be implemented by DAE as part of a strategic alliance with BADC as Grantee. The MoU between DAE and BADC will provide Grant Transfer from DAE to BADC, enabling BADC to carry out the activities. All five implementing agencies also contribute to their respective capacity building, technical assistance, and project management activities under Component 5. Each PIU will have separate fiduciary, environmental and social staff. Project Operational Manuals (POMs) will lay out roles, responsibilities, fiduciary, monitoring, and reporting arrangements for the project.
- 47. Project Steering Committee (PSC):** Each IA will have a PSC, chaired by the respective Senior Secretary/Secretary. Following the Government's Rules of Business, the PIUs will report to their respective PSC. The PSC for LGED will be chaired under the LGD, MoLGRD&C. The PSC for BWDB will be chaired under the MoWR. The PSC for DAE and BADC will be chaired under the MoA. The PSC for SDF will be chaired under the FID, MoF. The PSC for DDM will be chaired under MoDMR. The PSC will include representatives from ministries, divisions, departments/agencies that are part of overall implementation, coordination and strategy. The PSC will be responsible for: (i) providing implementation advice and operational guidance; (ii) reviewing financial and physical progress; (iii) resolving implementation problems; and (iv) providing any other necessary direction for effective implementation.



- 48. Project Implementation Committee (PIC):** Each IA will have a PIC, chaired by the head of agency (Chief Engineer of LGED, Director Generals of BWDB, DDM, DAE, and Managing Director of SDF), which will assist in the implementation and supervision of the respective components. The PIC is expected to include relevant representatives from ministries, divisions, departments/agencies. The PIC will ensure that implementation follows both Government and Bank rules and regulations. Specifically, the PIC will be responsible for: (i) supervising and reviewing implementation and providing necessary advice for timely delivery; (ii) monitoring and evaluating implementation progress and suggesting necessary course corrections; (iii) resolving issues and bottlenecks that may emerge during implementation; (iv) facilitation coordination and convergence with other line ministries, divisions, and departments/agencies; and (v) keeping the PSC apprised on overall performance and key issues relating to the project.
- 49. Each agency will have a dedicated Project Director and Deputy Project Director who will be responsible for managing project activities with the support of the PIU.**

B. Results Monitoring, Evaluation, and Verification Arrangements

- 50. LGED will be responsible for Monitoring and Evaluation (M&E) reporting of all result indicators to streamline M&E across Component 1 and 2. DDM and DAE will be responsible for Component 3 and 4, respectively.** All IAs will: (i) collect and report on project performance data (including physical and financial progress), and (ii) provide periodic information on intermediate project results and progress toward outcomes. Relevant line departments will assist the IAs by providing information. Throughout implementation, the IAs will prepare bi-annual progress reports. The project will finance gathering of baseline data to assess social, environmental, and economic impacts of key activities. The project outcomes and impacts will be evaluated through outcomes and intermediate level indicators as defined in the project's results framework. Detailed results M&E arrangements are elaborated in Annex 1.

C. Disbursement Arrangements

- 51.** Each PIU will open a Designated Account (DA) and be responsible for submitting disbursement/replenishment applications to the World Bank. The flow of funds will be through the five DAs of LGED, BWDB, DAE, DDM and SDF. BADC will receive fund from DAE as grant. Disbursements would initially be made on an advance and replenishment method. The allocation for DAE and BADC will be earmarked in a separate disbursement category in the Disbursement and Financial Information Letter. Other disbursement methods may include direct payment and special commitment. The Bank will disburse funds as per the UN Agreement with the GoB in case where the UN agencies are the service providers, based on issuance of UN commitment. The detailed arrangement is provided under Fiduciary section.

IV. PROJECT APPRAISAL SUMMARY

A. Technical, Economic and Financial Analysis

Technical analysis

- 52. Basic infrastructure rehabilitation and construction under Subcomponent 1.1 will use existing designs and standards based on the experience of LGED.** The IA has been implementing similar infrastructure projects, including those funded by the World Bank, such as the Emergency Multi-Sector Rohingya Crisis Response Project (P167762) and the Resilient Infrastructure for Adaptation and Vulnerability Reduction project (P173312). This approach ensures effective implementation while enhancing community resilience through improved climate and disaster-resilient design standards.



- 53. Selection and design of flood protection infrastructure (Subcomponent 1.2) will take into account several factors,** including level of damage from floods, future climate change risk, potential beneficiaries, land acquisition challenges, and availability of construction materials and contractors. Drawing from lessons learned in CEIP-I, investing in embankment protective works may prove to be a more cost-effective and timely approach to address riverbank erosion compared to retiring embankments. By fortifying existing riverbanks, the project can avoid the need for land acquisition. Furthermore, the project aims to balance the most urgent interventions essential for sustaining daily life, and most cost-effective works which bring higher benefits at lower costs.
- 54. Component 2 builds on a proven approach to livelihood improvement in Bangladesh.**⁴⁸ The strong community engagement fostered in predecessor projects have been instrumental in sustaining positive project outcomes. The project addresses significant gender gaps in disaster-affected communities, with a strong focus on women's empowerment through enhanced livelihood and entrepreneurship outcomes. The interventions ensure that up to 90 percent of participants in all processes and activities are women and will promote women's leadership and decision making. The component will tailor interventions to their and communities' specific needs and ensure sustainability of the interventions.
- 55. Component 3 builds on the lessons learned from several Bank financed projects**⁴⁹ and the longstanding experience of DDM as a mandated government agency to respond to disasters. Community engagement through public works and/or temporary livelihood opportunities has played a pivotal role in building community resilience and sustaining positive project outcomes in predecessor projects.
- 56. Component 4 applies the lessons learned from completed and ongoing Bank-financed projects and the longstanding experience of the two mandated government agencies.**⁵⁰ The production models and list of technologies developed and promoted under these projects have the potential to contribute to further increases in production and productivity. A uniform extension approach to mobilize individuals into groups is another area being considered.
- 57. The interventions are designed to build emergency preparedness and response (EP&R) capacity of IAs and beneficiaries.** Components 1 and 3 will strengthen EP&R capacity of government agencies through search and rescue boats and equipment, Emergency Operation Systems, and capacity building activities. Component 2 will enhance village-level EP&R capacity by establishing a village disaster management committee and provide training.
- 58. The project will support job creation. Components 1 and 4 will generate construction jobs for the local market, helping flood-affected workers find quality employment.** Components 2 and 3 will create jobs for the vulnerable population including youth and women who have not engaged in income-generating activities before. This access to jobs is crucial to reduce pressures to force migration in search of opportunities and helps restore economic status to pre-disaster levels and beyond. Additionally, Component 4 will offer 1.2 million person-days of short-term agricultural labor and staff opportunities for the selected districts.
- 59. Climate: The operation is aligned with the goals of the Paris Agreement on both adaptation and mitigation.**

⁴⁸Resilience, Entrepreneurship, and Livelihood Improvement Project (RELI) (P175820), the NJLIP (P149605), and the Social Investment Program Project I & II

⁴⁹ The NSNP project and the Inclusive Service and Opportunities for Host Communities and Displaced Rohingya Population project.

⁵⁰ Bangladesh Integrated Agricultural Productivity Project, NATP Phase I & II, Program on Agricultural and Rural Transformation for Nutrition, Entrepreneurship, and Resilience, Technical Assistance for Repurposing Agricultural Public Support Towards a Sustainable Food System Transformation in Bangladesh and Climate Smart Agriculture and Water Management Project



- 60. The project is consistent with the Bangladesh's Updated Nationally Determined Contribution (NDC) 2021⁵¹ and National Adaptation Plan (NAP) for 2030-50.⁵²** The project contributes to the implementation of the following NDC priorities: (a) improving road quality and traffic congestion; (b) reducing emission from upscaling alternate wetting and drying, improvement of fertilizer management, and replacement of low-productive animal with high producing crossbred cattle; and (c) supporting solar powered irrigation systems. The project contributes to the implementation of the NAP around the following areas: constructing and rehabilitating flood shelters; strengthening flood early warning system; installing lightning protection systems; strengthening search and rescue operations and emergency response equipment aids to develop an effective, well responsive disaster preparedness services; extension of climate-smart technologies for increasing irrigation water efficiently; augmentation of surface water for irrigation and multipurpose use; and extending stress tolerant, pest and disease-resistant rice and non-rice crops; crop diversification; and ensuring protection of women, socially disadvantaged and other vulnerable groups. Furthermore, the operation is consistent with Bangladesh's Country Climate and Development Report (CCDR). The CCDR noted that locally adapted and locally led solutions will be essential to building resilience, which will be embedded in the proposed Project design. The Project is also aligned with the South Asia Roadmap to implement World Bank's Climate Change Action Plan (CCAP) for 2021–2025. Among the five priority themes identified for Bangladesh, the proposed Project contributes to (1) adaptive delta management and coastal resilience and (4) low-carbon and resilient infrastructure.
- 61. Adaptation Risk Assessment and Reduction:** The project has been screened for climate change impacts and disasters. The overall exposure to extreme precipitation, flood, storm surge, lightning, and landslides is considered high in the project locations. Given that climate change may increase future risks posed by such hazards, the project took into consideration climate-resilient design features to improve access to basic services as well as increase the adaptive capacity of the government. Specifically, the project will establish climate-resilient all-weather roads and disaster shelters; rehabilitate flood embankment, hydraulic structures, and protective works; increase capacity of canals for stormwater drainage; promote climate smart agriculture, fisheries, and livestock practices; improve emergency response capacity for FSCD; rehabilitate and rebuild the damaged irrigation infrastructure; and cash transfers for vulnerable community members affected by flood. Overall, the project will equip agencies with adaptive measures and enhance the capacity of GoB to plan and coordinate response during current and future disasters. These measures will reduce material physical climate risks to an acceptable level.
- 62. Mitigation Risk Assessment and Reduction:** The project is not at a material risk of hindering the country's transition to a low greenhouse gas (GHG) emissions pathway. For example, the project will support crop production using Climate-Smart Agriculture (CSA) approaches⁵³; capacity building of farmers; flood management and protection; social protection, and cash transfer schemes for flood affected population; which are Universally Aligned as per the Multilateral Development Bank (MDB) List of Universally Aligned (UAL) activities (among other activities). Furthermore, the project will support the activities with low mitigation risk; for example: emergency response activities with construction of an Emergency Operations Center for strategic coordination and decision-making capacities during natural disasters, extreme climate events and emergencies as well as supply of fire safety and search and rescue equipment; all the buildings will be designed to be BNBC 2020 compliant on energy efficiency;⁵⁴ rehabilitation of rural roads with minor enhancement of capacity for all weather accessibility to rural community.⁵⁵ Though the remaining activities of components 1, 2, 3 and 4 are not in the MDB UAL, there are currently no

⁵¹ Ministry of Environment, Forest and Climate Change, GoB. 2021. *Bangladesh Nationally Determined Contributions 2021 (Updated)*.

⁵² Ministry of Environment, Forest and Climate Change, GoB. 2022. *National Adaptation Plan of Bangladesh (2023-2050)*.

⁵³ Contribute to resilience to climate change, and increased agricultural productivity and incomes.

⁵⁴ Refer to footnote 33 regarding BNBC energy efficiency standards.

⁵⁵ These roads will not create barriers to non-motorized traffic nor long-distance public transport as well as contribute to urban sprawl and cannot be substituted by lower transport modes such as railways or inland waterways.



alternative means to achieve the Development Objectives with lower GHG emissions. These activities are expected to lower GHG emissions and support efforts to reduce its carbon footprint using nature-based solutions in construction process.

- 63. Component 1 integrates climate resilience through:** (i) enhanced design practices using climate forecasts and probabilistic methods instead of solely historical data, ensuring infrastructure meets requirements under relevant climate scenarios; (ii) strategic implementation of hydraulic and drainage structures in flood-prone areas to prevent road deterioration; (iii) elevation of flood protection embankments based on climate risk assessments; and (iv) rehabilitation of flood-damaged road segments, drainage structures, bridges, and culverts adapted to projected hydrological changes and flooding intensity. The project incorporates future maintenance costs and adopts climate-resilient design standards capable of withstanding increased rainfall frequency and intensity under applicable climate scenarios, ensuring long-term sustainability.
- 64. Component 2 will enhance village-level disaster and climate resilience.** Research posits the importance of financial capital, assets, and networks as foundations for resilience to disasters.^{56,57} Under Subcomponent 2.1, Cash transfers will be provided to support the most vulnerable to meet immediate needs of flood recovery. Livelihood facilities, including shared facilities for agriculture, fisheries and livestock (e.g., existing livestock sheds) will be rehabilitated and improved with climate resilient designs, such as elevated platforms above the design flood levels and resilient structure capable of withstanding high winds. Community's awareness and readiness for disasters will be enhanced through various behavioral change campaigns (e.g., about emergency response protocols, emergency communication mechanisms, and evacuation plans during extreme climate events). Under Subcomponent 2.2, sub-loans and business development support will promote climate-smart IGAs. Beneficiaries will be provided technical guidance on investing in climate-smart agriculture, livestock, fisheries, and non-farm enterprises, including solar-powered irrigation systems, lithium powered community transport with affordable and energy-efficient swap stations, and organic alternatives to synthetic fertilizer and pest control to reduce GHG emission.
- 65. Component 3 will support short term livelihoods cash transfers (i.e., wages under EGPP+) for climate-vulnerable communities affected by the flood.** Additionally, the project will enhance community disaster preparedness through the provision of rescue boats to facilitate faster evacuation during floods, reduce casualties, and enable rapid delivery of essential supplies to stranded communities. The training, exercises, and drills (TEDs) program will build critical climate resilience capacities by training communities, volunteers, and local authorities in emergency operation planning, emergency operation center management, and climate-resilient infrastructure construction.
- 66. Component 4 will build disaster and climate resilience, restore the environmental balance, and enhance agricultural diversification.** Subcomponent 4.1 implements climate resilience through: (i) forming farmers groups that educate communities on climate change adaptation while promoting sustainable agriculture; (ii) introducing extreme climate event stress-tolerant, high-yielding, short-duration, and submergence-resistant crop varieties, particularly for rice, alongside climate-smart technologies; (iii) organizing field days showcasing modern production methods, balanced fertilizer application, climate-resilient crop varieties, and water-efficient irrigation systems (Alternate Wetting and Drying, drip irrigation); (iv) promoting vermi-compost and tricho-compost production for improved soil health management; and (v) establishing polyshed houses that optimize sunlight, water, and nutrient usage to produce quality seeds and seedlings with enhanced crop health, productivity, and resilience to climate stressors. Subcomponent 4.2 will address climate vulnerabilities through comprehensive irrigation infrastructure

⁵⁶ Azad, M. & Pritchard, B.. 2022. "Financial capital as a shaper of households' adaptive capabilities to flood risk in northern Bangladesh." *Ecological Economics* 195.

⁵⁷ Islam R., Walkerden, G. 2014. "How bonding and bridging networks contribute to disaster resilience and recovery on the Bangladeshi coast." *International Journal of Disaster Risk Reduction* 10 (Part A).



rehabilitation, including drainage network reconstruction, water storage enhancement, and implementation of flood-resilient technologies—all designed to withstand extreme precipitation, improve water use efficiency, and protect agricultural inputs against intensifying climate hazards.

67. Gender: Stark gender disparities exist in leadership and decision-making in Bangladesh, particularly in local government. Despite reserved seats for women at all levels of government, at both the Upazila and Union Parishad level, women comprise less than one percent of chairman positions.⁵⁸ Notably, women chairs of the Union Parishad nearly doubled between 2016 and 2021 (from 0.6 percent to 1 percent). Despite improvements and reserved seats, women face barriers to exercising their agency in their positions. Subcomponent 2.1 empowers women to make decisions in local decision-making by strengthening the role of community institutions which will comprise 90 percent female representation and be led by women with the leadership and capacity training provided. Gender gaps are also stark in livelihood opportunities. Although female labor force participation in rural areas increased from 39 percent in 2017 to 51 percent in 2022, it remains far below that of males at 80 percent; and women comprise about 30 percent of the labor force. Informal employment is also higher among women at 97 percent compared to 78 percent for men (2022) and approximately 29 percent of female employment is as “contributing family workers” compared to only 3 percent of male employment. In addition, women and youth face barriers in engaging productive livelihood and entrepreneurship activities, including (i) lack of business management or technical skills; (ii) lack of access to capital, finance, productive resources, and technology; (iii) limited networks; (iv) social and cultural barriers; and (v) mobility constraints. While there are services provided by the government and non-government organizations, these are often limited in scope, focusing on single interventions, such as skills development or access to finance. Subcomponent 2.2 provides an opportunity to address gender gaps in livelihood opportunities and increase own-account or wage-employed women by supporting them to (i) acquire skills for improved economic opportunities; (ii) loans to support income generation activities; and (iii) market linkage and business promotion support. Under sub-component 3.1, interventions will target and support short-term employment in public works and community services (through EGPP+) for women, with a target of 35 percent of total direct beneficiaries. Sub-component 4.1 will employ an inclusive beneficiary selection process targeting 30 percent women and 20 percent.

68. Citizen Engagement: The project implementation strategy will be embedded in a robust citizen engagement approach with multiple tools used for this purpose. The project Grievance Redress Mechanism (GRM) will be tailored to receive citizen’s feedback, and the resolution of the grievance cases will be actively followed up. Component 2 with SDF will involve the beneficiaries and communities in the project monitoring and social audit following the model already established under the ongoing Resilience, Entrepreneurship, and Livelihood Improvement (RELI) project. The project will also conduct user/beneficiary satisfaction surveys comprising of baselines, mid-lines and end-lines. These will be also included in the project’s results framework (i.e., community monitoring and social audit; and user/beneficiary satisfaction survey).

Economic and financial analysis

69. An economic analysis was conducted to evaluate the potential impacts of the proposed interventions. All components except Components 5 and 6 were included in the analysis.

70. The project is expected to have a positive net economic benefit. Considering an up to 30-year period and a discount rate of 10 percent, the economic net benefit is positive with a Net Present Value (NPV) of US\$165.9 million and an

⁵⁸ Data for Union Parishad is for 2021-2022 elections: 45 women elected as chairpersons out of approximately 4,578, a notable increase from 29 in the 2016 election. 3 out of 12 seats in each Union Parishad are reserved for women. For Upazila Parishad, the 2024 elections have a fifth phase which has yet to be completed - thus, data from 2019 elections is utilized, wherein 5 women were elected as chairpersons out of 489 upazilas; 1 of 2 vice-chairs and one-third of seats are reserved for women in each Upazila Parishad.



Internal Rate of Return (IRR) of 20.1 percent. A sensitivity analysis was conducted by increasing the capital expenditure and reducing the expected benefits (Table 2). The results show that the project will still produce net economic benefits despite such scenarios (i.e., 20 percent increase in cost and 20 percent decrease in benefits). This signifies the project's robustness to deliver positive economic outcomes.⁵⁹

Table 2 Summary of economic cost-benefit analysis

Scenario	Change	NPV (US\$ million)	IRR (percent)
Base case (all components)	N/A	165.9	20.1
Base case (subcomponent 1.1)	N/A	72.0	18.3
Base case (subcomponent 1.2)	N/A	31.9	16.2
Base case (subcomponent 2.1 & 2.2)	N/A	46.8	31.0
Component 3	N/A	2.42	34.8
Component 4 ⁶⁰	N/A	19.89	20.61
Increase in capital costs	20 percent	128.6	17.4
Decrease in benefits	20 percent	96.4	16.7
Increased cost + decreased benefits	20 percent	66.6	13.9

- 71. For subcomponent 1.1, the analysis focuses on the incremental benefits from better protection through improved shelters and improved connectivity due to climate-resilient roads.** Improved public spaces and schools are not monetized as part of this analysis but provide important benefits that are difficult to quantify. Including all costs, including US\$8.5 million for parts of this component not explicitly incorporated in this analysis, the NPV is US\$72 million and the IRR is 18.3 percent.
- 72. Subcomponent 1.2 rehabilitates flood protection infrastructure and aims to reduce the effects of future flood events.** To provide an initial estimate of the benefits of improved protective works, rehabilitation and flood preparedness, we assume that the recent floods were a 25-year return period event, and that the damages of such an event would be reduced by 15 percent due to the investments. This analysis implements a “risk to wellbeing” rather than a “risk to assets” approach based on the World Bank’s ‘Unbreakable’ microsimulation approach,⁶¹ to better incorporate the risks to these vulnerable households. Using a discount rate of 10 percent, a three percent growth of the benefits, and a 30-year project period, this yields an NPV of US\$31.9 million and an IRR of 16.2 percent.
- 73. Subcomponents 2.1 and 2.2 aim to empower local communities and institutions and enhance sustainable livelihoods by improving access to capital and employment.** The analysis examines three main elements: lending to small-scale entrepreneurs through the village credit organizations, employment generation, and incremental benefits from improved market and business linkages. To determine the returns from improved access to capital we use the detailed sector-level data on returns, and the distribution of types of investments made across sectors, collected as part of the economic analysis of the NJLIP (P149605). Based on these assumptions, the economic analysis yields a NPV of US\$46.8 million and an IRR of 31.0 percent.
- 74. Safety nets and livelihoods programs under Component 3 will boost consumption, income, and resilience at the household level.** Participation in the EGPP (about US\$2 per person per day) was found to boost overall household

⁵⁹ For components 2, 3 and 4 we use respectively a 15-year, 15-year and 5-year period.

⁶⁰ The MoA-FAO feasibility study conducted a detailed **economic analysis of the proposed interventions with the three key assumptions a)** Immediate interventions prevent long-term productivity losses, **b)** Recovery efforts are implemented efficiently with minimal delays, **and c)** External support supplements local funding gaps.

⁶¹ Hallegatte, S., Vogt-Schilb, A., Bangalore, M., & Rozenberg, J. 2016. *Unbreakable: building the resilience of the poor in the face of natural disasters*. World Bank Publications.



consumption by 7.7 percent to 10.6 percent on average.⁶² EGPP participation has been found to have other positive impacts on household welfare, such as improving food security, increasing average expenditure on healthcare services, and reducing outstanding household loans. With the increased amounts in transfers (about US\$3.5 per person per day) under the shock-responsive EGPP+ window, the consumption impacts are expected to be at least twice as great. Using the lower-bound estimate (consumption per capita of households in the bottom 40 percent), the net present value of cash transfer benefits is estimated at US\$200 (BDT 24,456) per beneficiary over a five-year period. Accordingly, the total NPV stands at US\$2.42 million, and the IRR is 34.8 percent.

- 75. A separate economic analysis⁶³ was conducted for component 4 to evaluate the potential impacts of the proposed agriculture interventions.** The project is expected to have a positive net economic benefit to the farmers with the adoption of project-promoted improved production technologies and irrigation support. Considering 15 years and a discount rate of 12 percent, the economic net benefit is positive with a NPV of US\$19.89 million, an IRR of 20.61 percent and the benefit-cost ratio is 1.85. The use of promoted improved technologies will contribute to 2 percent production increase, 4 percent yield gain, 2 percent labor and other inputs saving.
- 76. The economic analysis above demonstrates that the project yields a substantial positive internal rate of return and net present value,** suggesting that the proposed interventions are economically viable and beneficial. This analysis does not incorporate substantial other benefits, including to health, education and mental wellbeing which are more difficult to quantify but likely to be of significant importance to the project's impact.

B. Fiduciary

Financial Management

- 77. The overall financial management (FM) risk is “moderate” given the IAs’ adequate experience in managing IDA financed on-going and completed projects.** However, considering the complexity of the project, including dispersed operation and cash transfers, PIUs will implement actions to improve capacity and internal controls. In case of procuring a UN agency, for category 9 of the withdrawal table, UN commitment payment will be made as per the standard form of agreement.
- 78. FM risks mitigation measures include, among others,** (a) timely recruitment of FM specialist and training provided to the PIUs; (b) mainstreaming the Special Account Management Portal (i.e., Integrated Budgeting and Accounting System (iBAS++)) for processing all project transactions, accounting, financial reporting, and monitoring execution; (c) carrying out periodic internal audits to assess operational efficiency and to strengthen the internal control environment through the implementation of an agreed action plan; (d) using the Mandatory Direct Payment (MDP) would be applicable for this project and all payments relating to contracts under international open or limited competition or Direct Selection procurement as identified in the procurement plan would be made as direct payment or special commitment without paying from designated account or recipient's resources, regular direct payment and special commitment mechanism to make large US dollar payments; (e) direct transfer of wages/cash transfer/stipend to beneficiaries using banking channel/electronic transfer/mobile transfer; (f) arranging for periodic bipartite/tripartite meetings involving the project, the ministry, and the external auditors (Foreign Aided Project Audit Directorate [FAPAD]); and (g) establishing audit committees to follow up on the implementation of audit recommendations and to provide appropriate clarifications on external audit reports. The detailed FM arrangements and mitigation measures will be spelled out in the FM section of POM.

⁶² Cho and Ruthbah. 2018. Does Workfare Work Well?: The Case of the Employment Generation Program for the Poorest in Bangladesh. *Policy Research Working Paper*. No.8395. World Bank, Washington, DC.

⁶³ The analysis was conducted by DAE as part of preparation of the DPP. Detail table with the analysis is attached with Appendix 15 of the DPP.



- 79. Planning and budgeting:** An FM specialist and an accounts officer will carry out day-to-day FM functions in each PIU. Expenditures will be segregated as specified in the Government's Development Project Proposal (DPP) for LGED, BWDB, DDM, DAE, and BADC. SDF will follow non-ADP scheme for planning process. Budgeting and execution will utilize iBAS++, ensuring timely fund releases for project activities per budget allocation. Counterpart funding contribution will flow to the project as parallel financing. The project will use separate fund codes (Budget and Accounting Classification System) for each financing source, earmarking expenditures to avoid double-dipping during contractor payment approvals.
- 80. Eligibility of financing:** As per country financing parameters, IDA financing will not apply to specified categories of expenditures, including (a) costs associated with land and land use rights for the purpose of the project; (b) procurement of vehicles (except specialized and special purpose vehicles for the project as per procurement plan)⁶⁴; (c) recurrent expenditures for fuel, workshop allowances, sitting allowances, cash per diems, and honoraria, and salaries of civil servants, and (d) taxes exceeding 15 percent of the total IDA credit. The DPP will allocate counterpart funds as parallel financing to cover such excluded expenditure.
- 81. Funds flow and disbursement:** A DA, for example, in the form of Convertible Taka Special Account, will be opened for all PIUs with a national commercial bank to receive IDA funds. Disbursements would initially be made on an advance and replenishment method, based on regular statements of expenditure. DA ceiling will be fixed based on the fund requirement for two quarters. However, for large foreign contracts, the PIUs may use the special commitment and direct payment options. Financing of wages under EGPP+ would only be eligible for expenditures made from July 1, 2025, as documented in iBAS++ reports for program code (120001922000000) and economic code (3211134). Wages to EGPP+ beneficiaries will be paid through mobile financial services (MFS) and beneficiary details will be mentioned in the cash transfer record sheet. Expenditures under these payments will be recognized on receipt of utilization reports and when the funds are transferred to eligible beneficiaries. The MFS operator will confirm transactions to the PIU. These would be reimbursed to the Treasury. BADC will receive funds from DAE as grant and will maintain records and accounts adequately to reflect, in accordance with sound accounting practices, the operations, resources and expenditures incurred. BADC's expenditures would be included in the Statement of Expenditures (SOE). SDF may avail retroactive financing of US\$1 million (seven hundred fifty thousand Special Drawing Rights (SDR) 750,000) under the category Incremental Operating Cost (IOC) which will be subject fiduciary review before disbursement, and standard audit trail will be maintained by SDF.
- 82. For Component 2, SDF will transfer funds from its DA to its regional office operating accounts, which will, in turn, transfer to the community institution accounts.** Transfers from the community institution accounts to beneficiaries, including cash transfers and IGA loans, will be made directly to recipients' bank or mobile account. To ensure the transfer of funds and to preserve audit evidence, a copy of the national identify card of each recipient, along with respective mobile phone numbers, and acknowledgement of receipt of fund must be preserved at the project office. Community institutions will maintain records of all incurred and paid expenditures including sub loans and grants, which will be included in the SOE prepared by SDF. A robust monitoring framework will be established to ensure funds are utilized effectively, including (i) periodic inspection of community level documentation by SDF cluster officers and periodic reviews of SOEs and audit trail records by SDF regional and head offices. Incurred and paid expenditures by community institutions will be included in the SDF SOE. The community fiduciary responsibilities will be stipulated explicitly in the project participation agreement between community institutions and SDF. Detailed guidelines on reporting and fiduciary safeguards using the web-based loan management system and use of blockchain will be mentioned in the COM.

⁶⁴ A Country Financing Parameters exception has been granted for financing specialized and special purpose vehicles for the project based on Country Director email clearance dated December 18, 2024 and April 5, 2025.



- 83. Financial reporting:** Each PIU will prepare an Interim Unaudited Financial Report on a quarterly basis in the format agreed with the World Bank and submit it to the World Bank for its review and clearance within 45 days from the end of each quarter. The project will use iBAS++ for processing all transactions, accounting, financial reporting, and monitoring of execution.
- 84. Audit:** For the external audit, the FAPAD will express an opinion on the project financial statements in accordance with International Standards of Auditing and submit the report within six months of the end of each financial year. In addition, the auditor is required to provide a management letter containing the auditor's observations on internal controls and compliance with the financial covenants as laid out in the Financing Agreement. There is no pending audit report from any of the PIU for any of the projects funded by IDA. Regarding the internal audit, the project will engage a private auditor to conduct at least two rounds of internal audit over the project life to evaluate the project's internal control functions and operational effectiveness. It will prepare an action plan based on the recommendations of the internal auditor and monitor its implementation. The terms of reference for the internal audit function will be shared with the World Bank for concurrence before the engagement of the auditor.

Procurement

- 85. Procurement will be carried out in accordance with the World Bank Procurement Regulations for IPF Borrowers, Sixth Edition, February 2025 ("Procurement Regulations") and the provisions stipulated in the Project Procurement Strategy for Development (PPSD) and the procurement plan that may be updated with the World Bank's approval.** The project will be subject to the World Bank's Anticorruption Guidelines, dated October 15, 2006, and revised in January 2011 and July 2016. The project will use the World Bank's Systematic Tracking of Exchanges in Procurement (STEP) to plan, record, and track procurement transactions.
- 86. LGED, BWDB, SDF, DDM, DAE, and BADC will carry out procurement for their respective parts through the PIUs to be set up in each of the agencies.** Some procurement will be conducted by the respective field offices of the IAs. Total procurement expenditure of the Project is about US\$225 million. This will mainly involve procurement of goods, works, non-consulting and consultancy services to rehabilitate critical public infrastructure and to support affected households to recover from the economic impact of the floods. The agencies have been preparing PPCSD which spells out the detailed procurement arrangements, market approach, and identification of risks and mitigation measures. As part of the PPCSD, the IAs have prepared initial procurement plans specifying each contract to be financed under the project and the procurement method, market approach, estimated cost, established timelines for completion of the key milestones, World Bank's review requirements, etc. The procurement plan will be uploaded and maintained in STEP and updated at least annually or as and when necessary, during implementation. A General Procurement Notice will be published on the World Bank's website. Use of Rated Criteria will be mandatory for all procurement of goods, works and non-consulting services following an international market approach. As the project is prepared following Bank Policy for IPF Paragraph 12 and Guidance Note on Projects in situation of urgent need of assistance or capacity constraints, the project will exercise the available flexibility in the procurement activities. LGED, BWDB, and BADC will hire supervision consultants for their respective parts. Apart from supervision of the works, they will also act as "Project Manager/Engineer" for civil works contracts.
- 87. UN agencies would be selected when deemed appropriate.** DDM may engage a UN agency for the activities related to training, exercise and drills under component 3.2. UN agencies will only be selected for cases in which intervention by government and local implementing agencies may not be possible or when advantages are clear. Existing Bank instruments and standard forms of agreements will be used to engage relevant UN agencies. Agreement with any UN agency will be subject to prior review by the Bank.



- 88. Residual procurement risk after mitigation measures is assessed as ‘Substantial’ due to involvement of multiple IAs and emergency nature of activities.** Procurement risks include possible delays related to technical preparation, evaluation, and award processes. The risks will be mitigated through capacity building and technical assistance, combined with the IAs’ experience of previous similar projects funded by the World Bank, GoB, and other development partners. Other risk mitigation measures include hiring of additional technical and procurement experts on a needs-based approach, use of the government’s electronic procurement (e-GP) system, submission of quarterly procurement progress monitoring reports by each IA, providing weekly updates during bid evaluation after eight weeks of bid opening and during contract award after four weeks of Bid Evaluation Report approval, and fiduciary trainings for the relevant officials. The project shall follow the timeline for evaluation and contract award as per Bangladesh Public Procurement Rules – Schedule III- Part A [Rule 8(14)] – Procurement Processing and Approval Timetable – ensure the Notification of Appointment is issued within five weeks of Bank’s no-objection.
- 89. The World Bank’s procurement supervision will include prior and post review of the contracts.** Procurement post review by the Bank will be conducted on an annual basis or more frequently as needed.

C. Environmental, Social and Legal Operational Policies

- 90. The proposed project follows the Environmental and Social Framework (ESF).** The Project’s overall environmental and social impacts will be positive and contribute to the restoration of rural public infrastructure damaged by recent floods. Considering the type, scale of operation, nature, magnitude of potential ES risks and impacts from the proposed activities, the Project ES risk is rated as Substantial.
- 91. The Environmental risk is rated as Substantial.** The project primarily aims to rehabilitate and reconstruct infrastructure damaged by flooding during August-September 2024. The environmental risks anticipated from the project investments are rated substantial due to the emergency nature of the project design, limited knowledge about the exact scale and locations, and general implementation challenges. Key environmental risks include localized air and noise pollution, dust, waste generation, and pesticide use, along with community and occupational health and safety concerns. Although these impacts are limited in scope, they require effective management through environmental plans and adherence to occupational health and safety (OHS) protocols, in accordance with relevant standards of the ESF. Robust project management and stakeholder engagement will also be necessary to effectively address environmental risks.
- 92.** Any project intervention in any areas of environmental and ecological sensitivity and important ecosystem and with high environmental and social risk will be screened out during screening process. The ES screening will be required to evaluate if various interventions require specific environmental assessment. The ESMF will be prepared and disclosed within 60 days of project effectiveness to provide guidance about the generic ES impacts and mitigation plan, consultation and disclosure of various ES instruments. Following ESMF, site-specific screening/assessment to be conducted in accordance with the ESMF once the subproject is specifically designed and its location identified. To assess and mitigate the ES risks associated with the retroactively financed activities under the project, an Environmental and Social Audit, in line with ESS1, will be conducted by a third-party firm within 60-day of project effectiveness.
- 93. The social risk is rated as Substantial** due to the potential risks and impacts that may occur while implementing the rehabilitation, reconstruction and repair works associated with embankments, canals, roads, schools and shelters. The Project will use only government land and will lease or rent private land should it be needed for temporary storage of materials. No land acquisition will be required (or permitted) as all physical works will take place where



the infrastructure already exists. However, there are squatters present on the side of the embankment slopes who will be impacted by the rehabilitation works. All construction sites (once confirmed) will be screened for similar impacts which will be mitigated as per ESS5. The risk of labor influx is expected to be minimal as construction activities are expected to largely employ labor from local communities.

94. SDF activities will comprise of identifying vulnerable beneficiaries through a community-based approach and providing them with cash transfers, training, job facilitation, stipends, and access to a community-based revolving fund to finance their IGAs. This will follow the same model as the Bank-financed RELI project currently under implementation, including following the RELI ESMF for activities under retroactive financing. About US\$1 million SDF activities will be retroactively financed, comprising local-level institutional set-up, socioeconomic baseline survey, and possibly a small percentage of cash transfers. SDF has completed a rapid assessment of these activities, which are expected to carry low ES risks. An ES Audit will be carried out before disbursing funds for these activities. The Project ESMF will be in place within 60 days of effectiveness. SDF will screen for ES risks and apply appropriate mitigation measures, following ESF protocols and Bank guidance for all activities it undertakes.
95. SEA/SH (Sexual Exploitation and Abuse / Sexual Harassment) risks: Both civil works and safety net related activities can increase the risks of SEA/SH incidents. The SEA/SH risk for the Project is rated as Substantial using screening tools for civil works for Component 1, 3 and social protection for Component 2. Proportional mitigation measures will be laid out in the SEA/SH Mitigation Action Plan.
96. Consultation and communication as per ESS10 will be carried out for the Project as appropriate for the activities, locations and target communities. The required instruments, including the Stakeholder Engagement Plan (disclosed on April 4, 2025) and Environmental and Social Commitment Plan (disclosed on April 21, 2025), have been prepared and made publicly available on the IAS' and the World Bank websites⁶⁵. The Resettlement Policy Framework, Labor Management Plan and SEA/SH Mitigation Action Plan will be disclosed within 60 days of project effectiveness. The Grievance Redress Mechanism, including local committees will be established before the start of any Project works.

Legal Operational Policies	Triggered?
Projects on International Waterways OP 7.50	Yes
Projects in Disputed Area OP 7.60	No

97. OP 7.50 'Projects on International Waterways' is triggered for the Project as the activities will involve the potential use of the Ganges-Brahmaputra-Meghna River system and the Feni and Karnaphuli River systems. The exception to the notification requirement under paragraph 7 (a) applies because the rural infrastructure and flood control infrastructure focus on rehabilitating and upgrading existing schemes. The works will not adversely change the quality or quantity of water being used by other riparian countries and will not be adversely affected by other riparians' possible change in water use. The Regional Vice President approved on December 8, 2024 an exception to the notification requirement of riparian states under World Bank Policy OP7.50, based on the above considerations.

⁶⁵ Links to Environmental Social Commitment Plans: [ESCP LGED](#); [ESCP BWDB](#); [ESCP SDF](#); [ESCP DAE](#); [ESCP BADC](#); [ESCP DDM](#).
Links to Stakeholder Engagement Plans: [SEP LGED](#); [SEP BWDB](#); [SEP SDF](#); [SEP DAE](#); [SEP BADC](#); [SEP DDM](#).



98. In addition, the Regional Vice President approved on March 27, 2025, an exception under paragraph 7(a) of the World Bank Policy OP 7.50, applicable for restoring agricultural production under Component 4 which will focus on rehabilitating and upgrading existing schemes. Rebuilding irrigation and storage facilities, repair and reconstruction of damaged irrigation equipment and infrastructure, re-excavation of small water bodies and construction of buried pipelines will promote adoption of good agriculture practice standards and other climate-smart agriculture technologies leading to a reduction of the use of pesticides and fertilizers, which in turn, result in water quality improvements. The works will not adversely change the quality or quantity of water being used by other riparian countries and will not be adversely affected by other riparians' possible change in water use.

Grievance Redress Services:

99. Grievance Redress. Communities and individuals who believe that they are adversely affected by a project supported by the World Bank may submit complaints to existing project-level grievance mechanisms or the Bank's Grievance Redress Service (GRS). The GRS ensures that complaints received are promptly reviewed in order to address project-related concerns. Project affected communities and individuals may submit their complaint to the Bank's independent Accountability Mechanism (AM). The AM houses the Inspection Panel, which determines whether harm occurred, or could occur, as a result of Bank non-compliance with its policies and procedures, and the Dispute Resolution Service, which provides communities and borrowers with the opportunity to address complaints through dispute resolution. Complaints may be submitted to the AM at any time after concerns have been brought directly to the attention of Bank Management and after Management has been given an opportunity to respond. For information on how to submit complaints to the Bank's GRS, visit <http://www.worldbank.org/GRS>. For information on how to submit complaints to the Bank's Accountability Mechanism, visit <https://accountability.worldbank.org>.

V. KEY RISKS

100. **The overall project risk is rated as Substantial.** The Risk Categories rated Substantial are described below.

Table 4 Summary of key risks

1. Political and Governance (Substantial)
Risk: Bangladesh is currently led by an interim government since August 2024, and the political situation is anticipated to remain volatile during the transition to a newly elected government. Safety and security concerns persist, despite the priority placed by the interim government on law and order. The interim government is implementing critical reforms before holding elections.
Mitigation Measures: Maintaining dialogue with key stakeholders at all levels including government officials, local leaders, and community groups, to ensure their continued support and immediately address any emerging concerns.
2. Macroeconomic (Substantial)
Risk: Downside risks have increased, related to uncertainties in the political process and the security situation which impact the recovery of economic activities. The financial sector remains vulnerable due to deviations from international regulatory and supervisory standards, and weak corporate governance. The Balance of Payment remained in deficit and the foreign exchange reserves remained under pressure.
Mitigation measures: Partly mitigated by the GoB's reform program, supported by the IMF arrangements, ongoing World Bank and other DPs policy lending.
3. Institutional Capacity for Implementation and Sustainability (Substantial)
Risk: This is mainly due to the project's multisectoral complexity and the need for coordination among five PIUs for timely implementation. Though each IA has had projects supported by the World Bank before, it is the first time for the IAs to work on a same project simultaneously, especially in a disaster recovery setting which requires a higher degree of coordination, monitoring, and evaluation.



Mitigation measures: Partly mitigated by having LGED to be the main responsible agency for M&E reporting of all result indicators for LGED, BWDB, and SDF, to streamline the M&E for the noted three IAs which are responsible for 85 percent of the project's financing.

4. Fiduciary (Substantial)

Risk: This is mainly because of the procurement risks, due to the involvement of multiple IAs and emergency nature of the activities. This may induce possible delays related to technical preparation, evaluation, and award processes in the procurement activities.

Mitigation measures: Mitigation measures include hiring additional capacity and providing training to improve overall FM and procurement performance; streamlining procurement and disbursement processes to ensure efficiency and transparency; hiring of additional technical and procurement experts on a needs-based approach; use of the government's electronic procurement system; providing weekly updates during bid evaluation after eight weeks of bid opening and during contract award after four weeks of Bid Evaluation Report approval; utilizing integrated budget systems, maintain real-time accounting entries, and separate ledger codes for each source of financing; preparing quarterly interim un-audited financial reports to monitor expenditures; producing quarterly procurement progress monitoring reports to identify bottlenecks and recommend corrective measure; undertaking an annual risk-based internal audit by independent audit firm; managing residual risks through proactive and continuous supervision and dialogue with the GoB.

5. Environmental & Social (Substantial)

Risk: Key environmental risks include localized air and noise pollution, dust, waste generation, and pesticide use, along with community and occupational health and safety concerns. Key social risks are due to impacts that may occur while implementing the infrastructure rehabilitation, reconstruction and repair works.

Mitigation measures: Enforcing adherence to OHS protocols, in accordance with relevant standards of the ESF; ensuring robust project management and stakeholder engagement; carrying out an E&S Audit, in line with ESS1, by a third-party firm.



ANNEX 1. RESULTS FRAMEWORK

PDO Indicators by PDO Outcomes

Baseline	Closing Period
Increased climate and disaster resilience	
People benefitting from climate resilient infrastructure (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	1,662,109
➤ People benefitting from climate resilient infrastructure - Female (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	831,055
➤ People benefitting from climate resilient infrastructure - Youth (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	317,629
People benefitting from climate resilient planning, preparation, surveillance, and/or response (Number of people) ^{CRI}	
Apr/2025	Dec/2030
81,000	413,547
➤ People benefitting from climate resilient planning, preparation, surveillance, and/or response - Female (Number of people) ^{CRI}	
Apr/2025	Dec/2030
40,500	206,774
➤ People benefitting from climate resilient planning, preparation, surveillance, and/or response - Youth (Number of people) ^{CRI}	
Apr/2025	Dec/2030
15,479	79,029
➤ People benefitting from reduced exposure to climate-related hazards due to disaster shelters (Number of people)	
Apr/2025	Dec/2030
81,000	237,000
➤ People benefitting from flood risk reduction through rehabilitated flood protection infrastructure (Number of people)	
Apr/2025	Dec/2030
0	176,547



Strengthened livelihoods	
Beneficiaries of cash-based interventions (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	37,731
➤ Beneficiaries of cash-based interventions - Female (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	18,865
➤ Beneficiaries of cash-based interventions - Youth (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	7,169
Beneficiaries of economic inclusion programs (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	374,880
➤ Beneficiaries of economic inclusion programs - Female (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	187,440
➤ Beneficiaries of economic inclusion programs - Youth (Number of people) ^{CRI}	
Apr/2025	Dec/2030
0	71,227
People benefitting from access to improved agriculture production system (Number of people)	
Apr/2025	Dec/2030
0	65,000
➤ People benefitting from access to improved agriculture production system - Female (Number of people)	
Apr/2025	Dec/2030
0	19,500
➤ People benefitting from access to improved agriculture production system - Youth (Number of people)	
Apr/2025	Dec/2030
0	13,000

Intermediate Indicators by Components

Baseline	Closing Period
People Centric Resilient Infrastructure	



Climate resilient multipurpose disaster shelters/primary schools constructed (Number)	
Apr/2025	Dec/2030
0	25
Climate resilient multipurpose disaster shelters/primary schools rehabilitated and capacity enhanced (Number)	
Apr/2025	Dec/2030
0	54
Emergency evacuation bridge rehabilitated and reconstructed (Meter(m))	
Apr/2025	Dec/2030
0	1,150
Emergency evacuation rural roads rehabilitated (Kilometers)	
Apr/2025	Dec/2030
0	485
Lightning protection systems installed (Number)	
Apr/2025	Dec/2030
0	100
Emergency Operation Systems Installed (Yes/No)	
Apr/2025	Dec/2030
No	Yes
Search and rescue equipment supplied to FSCD (Yes/No)	
Apr/2025	Dec/2030
No	Yes
Canal re-excavated (Meter(m))	
Apr/2025	Dec/2030
0	19,600
Embankment re-sectioned and repaired (Meter(m))	
Mar/2025	Dec/2030
0	65,000
Slope protection works completed (Meter(m))	
Apr/2025	Dec/2030
0	2,560
Bank protection works completed (Meter(m))	
Apr/2025	Dec/2030
0	37,300
Hydraulic structures repaired (Number)	



Apr/2025	Dec/2030
0	21
Flood forecasting system enhanced (Yes/No)	
Apr/2025	Dec/2030
No	Yes
Beneficiaries aware of project interventions and engaged through outreach activities (Percentage)	
Apr/2025	Dec/2030
0	70
Livelihoods Recovery and Enhancement Support	
Beneficiaries wage or self employed as a result of project interventions (Number of people)	
Apr/2025	Dec/2030
0	44,220
➤ Beneficiaries wage or self employed as a result of project interventions - Female (Number of people)	
Apr/2025	Dec/2030
0	39,798
➤ Beneficiaries wage or self employed as a result of project interventions - Youth (Number of people)	
Apr/2025	Dec/2030
0	4,620
Share of beneficiaries with increased awareness of health, hygiene, nutrition, and disaster preparedness issues (Percentage)	
Apr/2025	Dec/2030
0	75
People provided cash transfer support (Number of people)	
Apr/2025	Dec/2030
0	88,000
Community institutions established with active Village Development Funds (Number)	
Apr/2025	Dec/2030
0	550
Cluster and Regional Community Societies established (Number)	
Apr/2025	Dec/2030
0	24
Share of women in community institutions' decision making and management positions (Percentage)	
Apr/2025	Dec/2030
0	90
Productive Alliances registered (Number)	



Apr/2025	Dec/2030
0	25
Share of beneficiaries who are satisfied with project activities (Percentage)	
Apr/2025	Dec/2030
0	70
People who receive loans for income generation activities (Number of people)	
Apr/2025	Dec/2030
0	66,000
People completed skills development training (Number of people)	
Apr/2025	Dec/2030
0	7,700
➤ People completed skills development training - Female (Number of people)	
Apr/2025	Dec/2030
0	3,500
Follow-up measures undertaken from recommendations of the social audit and community monitoring reports (Percentage)	
Apr/2025	Dec/2030
0	70
Community Recovery and Resilience to Disasters	
Work-days generated by short-term work opportunities (Number)	
Apr/2025	Dec/2030
0	3,188,520
➤ Work-days generated by short-term work opportunities - Female (Number)	
Apr/2025	Dec/2030
0	1,115,982
Search and rescue boats supplied to DDM (Number)	
Apr/2025	Dec/2030
0	370
Person-days of training provided to communities on SAR operation, DRM information and emergency preparedness (Number)	
Apr/2025	Dec/2030
0	14,500
Agricultural System Restoration	
Area covered under climate resilient technologies (Hectare(Ha))	
Apr/2025	Dec/2030
0	23,000



Productivity increase of rice (Percentage)	
Apr/2025	Dec/2030
0	4
Productivity increase of high-value crops (Percentage)	
Apr/2025	Dec/2030
0	5
Client days training conducted on crop and seed production, agricultural tools and machinery use, CSA technology and disaster risk reduction in agriculture (Days)	
Apr/2025	Dec/2030
0	100,000
Storage and service facilities reconstructed (Number)	
Mar/2025	Dec/2030
0	15
Area covered by improved irrigation facilities and networks (Hectare(Ha))	
Apr/2025	Dec/2030
0	6,470
Irrigation and drainage network renovated and reconstructed (Kilometers)	
Apr/2025	Dec/2030
0	306
Project Administration, Management, Coordination, Monitoring and Evaluation	
Grievance Redress Committee (GRC) established and functional (Number)	
Apr/2025	Dec/2030
0	18
Contingent Emergency Response	



Monitoring & Evaluation Plan: PDO Indicators by PDO Outcomes

Increased climate and disaster resilience	
People benefiting from climate resilient infrastructure (Number of people) ^{CRI}	
Description	Number of people with access to at least one climate resilient infrastructure as a result of this project: emergency evacuation rural roads rehabilitated, emergency evacuation bridge rehabilitated and reconstructed. Access refers to being able to utilize or benefit from the infrastructure – meaning that the infrastructure should be physically accessible, available, and affordable.
Frequency	Biannual
Data source	LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
People benefiting from climate resilient infrastructure - Female (Number of people) ^{CRI}	
Description	Number of female with access to at least one climate resilient infrastructure as a result of this project: emergency evacuation rural roads rehabilitated, emergency evacuation bridge rehabilitated and reconstructed.
Frequency	Biannual
Data source	LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
People benefiting from climate resilient infrastructure - Youth (Number of people) ^{CRI}	
Description	Number of youth with access to at least one climate resilient infrastructure as a result of this project: emergency evacuation rural roads rehabilitated, emergency evacuation bridge rehabilitated and reconstructed. Youth is defined as people within the age group of 15-24.
Frequency	Biannual
Data source	LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
People benefitting from climate resilient planning, preparation, surveillance, and/or response (Number of people) ^{CRI}	
Description	Number of people benefitting from flood risk reduction through the rehabilitation, repair and construction of flood protection structures as well as with reduced exposure to climate-related hazards due to disaster shelters.
Frequency	Biannual
Data source	BWDB and LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB and LGED
People benefitting from climate resilient planning, preparation, surveillance, and/or response - Female (Number of people) ^{CRI}	
Description	Number of female benefitting from flood risk reduction through the rehabilitation, repair and construction of flood protection structures as well as with reduced exposure to climate-related hazards due to disaster shelters.
Frequency	Biannual
Data source	BWDB and LGED administrative reports
Methodology for Data Collection	Report review and site visits



Responsibility for Data Collection	BWDB and LGED
People benefitting from climate resilient planning, preparation, surveillance, and/or response - Youth (Number of people) ^{CRI}	
Description	Number of youth benefitting from flood risk reduction through the rehabilitation, repair and construction of flood protection structures as well as with reduced exposure to climate-related hazards due to disaster shelters. Youth is defined as people within the age group of 15-24.
Frequency	Biannual
Data source	BWDB and LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB and LGED
People benefitting from reduced exposure to climate-related hazards due to disaster shelters (Number of people)	
Description	Number of people with access to improved climate resilient multipurpose disaster shelters, as measured by the design capacity, as a result of this project. Access refers to being able to utilize or benefit from the infrastructure – meaning that the infrastructure should be physically accessible, available, and affordable.
Frequency	Biannual
Data source	LGED administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
People benefitting from flood risk reduction through rehabilitated flood protection infrastructure (Number of people)	
Description	Number of people benefitting from flood risk reduction through the rehabilitation, repair and construction of flood protection structures.
Frequency	Biannual
Data source	BWDB administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB
Strengthened livelihoods	
Beneficiaries of cash-based interventions (Number of people) ^{CRI}	
Description	This indicator measures the number of individuals benefitting from social safety net programs (EGPP+) from the project. It includes the activities (e.g. cash transfers and/or public works) under Component 3. Data will be disaggregated by gender and age. The objective is to ensure poor and vulnerable households have work opportunities to generate income to build resilience while enhancing community assets (i.e., roads, embankments, etc.).
Frequency	Biannual
Data source	DDM's EGPP+ MIS
Methodology for Data Collection	MIS reports; site visits; spot check surveys
Responsibility for Data Collection	DDM
Beneficiaries of cash-based interventions – Female (Number of people) ^{CRI}	
Description	This indicator measures the number of female benefitting from social safety net programs (EGPP+) from the project. It includes the activities (e.g. cash transfers and/or public works) under Component 3. Data will be disaggregated by gender and age. The objective is to ensure poor and vulnerable households have work opportunities to generate income to build resilience while enhancing community assets (i.e., roads, embankments, etc.).



Frequency	Biannual
Data source	DDM's EGPP+ MIS
Methodology for Data Collection	MIS reports; site visits; spot check surveys
Responsibility for Data Collection	DDM
Beneficiaries of cash-based interventions – Youth (Number of people) ^{CRI}	
Description	This indicator measures the number of youth benefitting from from social safety net programs (EGPP+) from the project. It includes the activities (e.g. cash transfers and/or public works) under Component 3. Data will be disaggregated by gender and age. The objective is to ensure poor and vulnerable households have work opportunities to generate income to build resilience while enhancing community assets (i.e., roads, embankments, etc.). Youth is defined as people within the age group of 15-24.
Frequency	Biannual
Data source	DDM's EGPP+ MIS
Methodology for Data Collection	MIS reports; site visits; spot check surveys
Responsibility for Data Collection	DDM
Beneficiaries of economic inclusion programs (Number of people) ^{CRI}	
Description	This indicator measures the total number of people who are benefitted from social safety net programs from the project. It includes the activities (e.g. cash transfers, economic inclusion and/or income generation support) under Component 2.
Frequency	Biannual
Data source	Administrative data; SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Beneficiaries of economic inclusion programs - Female (Number of people) ^{CRI}	
Description	This indicator measures the total number of female who are benefitted from social safety net programs from the project. It includes the activities (e.g. cash transfers, economic inclusion and/or income generation support) under Component 2.
Frequency	Biannual
Data source	Administrative data; SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Beneficiaries of economic inclusion programs - Youth (Number of people) ^{CRI}	
Description	This indicator measures the total number of youth who are benefitted from social safety net programs from the project. It includes the activities (e.g. cash transfers, economic inclusion and/or income generation support) under Component 2. Youth is defined as people within the age group of 15-24.
Frequency	Bi-annual
Data source	Administrative data; SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS
Responsibility for Data Collection	SDF
People benefitting from access to improved agriculture production system (Number of people)	



Description	This indicator measures the number of individuals who received training, inputs and or any service from the project. It includes group members, trainees, and inputs recipients. People benefitting from the distributed farm machinery, irrigation and drainage network restoration and, storage facilities will be counted as indirect beneficiary. A person will be counted once. Access refers to being able to utilize or benefit from the production technologies, infrastructure and facilities created with project support.
Frequency	Biannual
Data source	DAE's and BADC's Internal MIS
Methodology for Data Collection	Activity records and project MIS
Responsibility for Data Collection	DAE and BADC Local Staff
People benefitting from access to improved agriculture production system - Female (Number of people)	
Description	At least 30 percent of the total beneficiaries is expected to be female.
Frequency	Biannual
Data source	DAE's and BADC's Internal MIS
Methodology for Data Collection	Activity records and project MIS
Responsibility for Data Collection	DAE and BADC Local Staff
People benefitting from access to improved agriculture production system - Youth (Number of people)	
Description	Youth is defined as people within the age group of 18-35 years. At least 20 percent of the total beneficiaries is expected to be youth.
Frequency	Biannual
Data source	DAE's and BADC's Internal MIS
Methodology for Data Collection	Activity records and project MIS
Responsibility for Data Collection	DAE and BADC Local Staff

Monitoring & Evaluation Plan: Intermediate Results Indicators by Components

People centric resilient infrastructure	
Climate resilient multipurpose disaster shelters/primary schools constructed (Number)	
Description	Number of climate resilient multi-purpose shelters/primary schools constructed (cumulative). Average design capacity per shelter is 3,000 people.
Frequency	Biannual
Data source	LGED Administrative Reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Climate resilient multipurpose disaster shelters/primary schools rehabilitated and capacity enhanced (Number)	
Description	Number of climate resilient multi-purpose shelters/primary schools rehabilitated and capacity enhanced (cumulative). Average design capacity per shelter will be 3000 people from 1500 people.
Frequency	Biannual
Data source	LGED Administrative Reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data	LGED



Collection	
Emergency evacuation bridge rehabilitated and reconstructed (Meter(m))	
Description	Length (meter) of emergency evacuation bridge and culvert rehabilitated and reconstructed.
Frequency	Biannual
Data source	LGED Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Emergency evacuation rural roads rehabilitated (Kilometers)	
Description	Length (kilometer) of all-weather climate resilient emergency evacuation rural road rehabilitated.
Frequency	Biannual
Data source	LGED Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Lightning protection systems installed (Number)	
Description	Number of lightning protection systems installed to reduce the risk of lightning strikes resulting from extreme hydro-meteorological events.
Frequency	Biannual
Data source	LGED Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Emergency Operation Systems Installed (Yes/No)	
Description	Emergency Operation Systems for upazilas installed as a result of the project.
Frequency	Biannual
Data source	LGED Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Search and rescue equipment supplied to FSCD (Yes/No)	
Description	FSCD facilities or warehouse equipped with sufficient search and rescue equipment as a result of the project.
Frequency	Biannual
Data source	LGED Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED
Canal re-excavated (Meter(m))	
Description	Length (meter) of canal re-excavated.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB



Embankment re-sectioned and repaired (Meter(m))	
Description	Length (meter) of embankment re-sectioned and repaired.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB
Slope protection works completed (Meter(m))	
Description	Length (meter) of slope protected.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB
Bank protection works completed (Meter(m))	
Description	Length (meter) of river bank protected.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB
Hydraulic structures repaired (Number)	
Description	Number of Hydraulic structures repaired.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	BWDB
Flood forecasting system enhanced (Yes/No)	
Description	This indicator measures the enhancement of the Flood Forecasting Model and/or System.
Frequency	Biannual
Data source	BWDB Administrative reports
Methodology for Data Collection	Report review and model output
Responsibility for Data Collection	BWDB
Beneficiaries aware of project interventions and engaged through outreach activities (Percentage)	
Description	This indicator measures the proportion of beneficiaries within the project area who are informed about the project's interventions (specifically targeting component 1, which involves resilient rural infrastructure and flood protection rehabilitation) and who have been actively engaged through various outreach and communication activities.
Frequency	Biannual
Data source	LGED and BWDB administrative reports
Methodology for Data Collection	- Surveys and feedback forms (pre and post-outreach activities). - Volunteer reports and outreach session attendance records. - Digital/social media engagement analytics.



Responsibility for Data Collection	LGED and BWDB
Livelihoods Recovery and Enhancement Support	
Beneficiaries wage or self employed as a result of project interventions (Number of people)	
Description	This indicator counts the number of individuals who have sustained employment due to the subcomponent 2.2. The target is based on the assumption that 60 percent of those who received loans and youth attended training will be wage or self employed.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Beneficiaries wage or self employed as a result of project interventions - Female (Number of people)	
Description	This indicator counts the number of female who have sustained employment due to the subcomponent 2.2. The target assumes 90 percent of the total beneficiaries wage or self-employed will be female.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Beneficiaries wage or self employed as a result of project interventions - Youth (Number of people)	
Description	This indicator counts the number of youth who have sustained employment due to the subcomponent 2.2. Youth is defined as people within the age group of 15-24. The target assumes 60 percent of the total trained youth will gain wage or self-employment.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Share of beneficiaries with increased awareness of health, hygiene, nutrition, and disaster preparedness issues (Percentage)	
Description	This indicator measures the share of people who have acquired knowledge by participating in health, hygiene, nutrition, and disaster preparedness awareness-raising initiatives funded by a project.
Frequency	Twice in project life
Data source	Survey
Methodology for Data Collection	Sample survey of those who participated in awareness raising campaign.
Responsibility for Data Collection	SDF
People provided cash transfer support (Number of people)	
Description	This indicator counts the number of individuals who receive cash transfers from the project.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Community institutions established with active Village Development Funds (Number)	



Description	This indicator measures the number of community institutions established and signed Umbrella Financing Agreement for VDF with SDF.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Cluster and Regional Community Societies established (Number)	
Description	This indicator measures the number of Cluster and Regional Community Societies established by SDF.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at cluster/district level through SDF's MIS.
Responsibility for Data Collection	SDF
Share of women in community institutions' decision making and management positions (Percentage)	
Description	This indicator measures the proportion of women in community institutions' decision making and management positions. Community-based decision making and management positions are defined in the COM. Women and affected and vulnerable are not mutually exclusive categories. Therefore, some women counted in this indicator may also be counted as the affected and vulnerable in community institutions' decision-making and management positions.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Productive Alliances registered (Number)	
Description	This indicator counts the number of Productive Alliances that have completed the legal registration process and are recognized as formal entities.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Share of beneficiaries who are satisfied with project activities (Percentage)	
Description	This indicator measures the percentage of beneficiaries who report being satisfied with the project activities they have participated in or received support from.
Frequency	Twice in project life
Data source	Survey
Methodology for Data Collection	Sample satisfaction survey
Responsibility for Data Collection	LGED, BWDB, and SDF
People who receive loans for income generation activities (Number of people)	
Description	This indicator counts the number of individuals who receive loans from the VDF for IGAs.
Frequency	Biannual
Data source	SDF's MIS



Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
People completed skills development training (Number of people)	
Description	This indicator counts the number of individuals who have finished a training program.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
People completed skills development training - Female (Number of people)	
Description	This indicator counts the number of female individuals who have finished a training program.
Frequency	Biannual
Data source	SDF's MIS
Methodology for Data Collection	Information collected at village level through SDF's MIS.
Responsibility for Data Collection	SDF
Follow-up measures undertaken from recommendations of the social audit and community monitoring reports (Percentage)	
Description	Observations submitted to village community institutions by Social Audit Committee (SAC) resolved
Frequency	Biannual
Data source	SDF's GRM and Reports by social audit and community monitoring committees
Methodology for Data Collection	Grievance reports to SDF's GRM and periodic social audit and community monitoring reports
Responsibility for Data Collection	SDF
Community Recovery and Resilience to Disasters	
Work-days generated by short-term work opportunities (Number)	
Description	This indicator measures the number of workdays generated under EGPP+. Data will be disaggregated by gender and age. The objective is to ensure poor and vulnerable households have access to dignified work income earning opportunities, and thus reduce reliance to negative coping strategies.
Frequency	Biannual
Data source	DDM's EGPP+ MIS
Methodology for Data Collection	MIS reports; site visits; spot check surveys
Responsibility for Data Collection	DDM
Work-days generated by short-term work opportunities - Female (Number)	
Description	This indicator measures the number of workdays generated under EGPP+ for female (female target is 35%). The objective is to ensure poor and vulnerable households have access to dignified work income earning opportunities, and thus reduce reliance to negative coping strategies.
Frequency	Biannual
Data source	DDM's EGPP+ MIS
Methodology for Data Collection	MIS reports; site visits; spot check surveys
Responsibility for Data Collection	DDM
Search and Rescue boats supplied to DDM (Number)	



Description	District Relief Godown facilities or warehouses equipped with number search and rescue boats equipment as a result of the project.
Frequency	Biannual
Data source	DDM Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	DDM
Person-days of training provided to communities on SAR operation, DRM information and emergency preparedness (Number)	
Description	This indicator counts the number of person days attended by community volunteers under the training activity.
Frequency	Biannual
Data source	DDM Administrative reports
Methodology for Data Collection	DDM progress reports, MIS
Responsibility for Data Collection	DDM
Agricultural System Restoration	
Area covered under climate resilient technologies (Hectare(ha))	
Description	Captures farmlands in hectares covered under different practices of the adopted farmers.
Frequency	Annual
Data source	Field monitoring reports, Survey reports
Methodology for Data Collection	Survey, Report review
Responsibility for Data Collection	DAE
Productivity increase of rice (Percentage)	
Description	This indicator measures average increases in yield of rice (Aus, Aman and Boro) in project areas. Baseline figure will be confirmed through baseline survey.
Frequency	Annual
Data source	Demo records, survey reports
Methodology for Data Collection	Demo records, Survey reports
Responsibility for Data Collection	DAE
Productivity increase of high-value crops (Percentage)	
Description	This indicator measures average increases in yield of high-value crops in project areas. Baseline survey will confirm the types of crops and baseline figure.
Frequency	Annual
Data source	Demo records, survey reports
Methodology for Data Collection	Demo records, Survey reports
Responsibility for Data Collection	DAE
Client days training conducted on crop and seed production, agricultural tools and machinery use, CSA technology and disaster risk reduction in agriculture (Days)	
Description	Client refers to farmers, machinery service receivers, and all other stakeholders who receive both formal and informal training from the project. It measures the number of client days of training provided, i.e., the number of clients who complete training multiplied by the duration of training expressed in days. The term 'trained' means any training organized or provided by the project (formal or informal training). The term 'day' may include aggregating partial day or hours to full client day.



Frequency	Biannual
Data source	DAE's training records in MIS
Methodology for Data Collection	Training and field days records
Responsibility for Data Collection	DAE
Storage and service facilities reconstructed (Number)	
Description	Number of fertilizer and seed storage godown constructed/rehabilitated.
Frequency	Biannual
Data source	BADC's Administrative and M&E reports
Methodology for Data Collection	Reports review and site visits
Responsibility for Data Collection	BADC
Area covered by improved irrigation facilities and networks (Hectare(ha))	
Description	Measures hectare coverage under the installed irrigation facilities and networks (new/repair).
Frequency	Biannual
Data source	BADC's Administrative and M&E reports
Methodology for Data Collection	Reports review and site visits
Responsibility for Data Collection	BADC
Irrigation and drainage network renovated and reconstructed (Kilometers)	
Description	It counts length (kilometer) of the small-scale water bodies (Khal/Nalas/Boro pit) excavated, the uPVC drainage and buried pipeline constructed/renovated in kilometer.
Frequency	Biannual
Data source	BADC's Administrative and M&E reports
Methodology for Data Collection	Reports review and site visits
Responsibility for Data Collection	BADC
Project Administration, Management, Coordination, Monitoring and Evaluation	
Grievance Redress Committee (GRC) established and functional (Number)	
Description	Number of GRCs formed, and functional with Grievance Redress Mechanism (GRM) accessible to the communities and affected persons as well as receiving and processing complaints, queries, suggestions, positive feedback, and concerns of project-affected parties related to the environmental and social performance of a project.
Frequency	Biannual
Data source	LGED, BWDB, SDF, DDM, DAE, and BADC Administrative reports
Methodology for Data Collection	Report review and site visits
Responsibility for Data Collection	LGED, BWDB, SDF, DDM, DAE, and BADC



ANNEX 2: ADDITIONAL INFORMATION ON PROJECT BACKGROUND AND CLIMATE IMPLICATIONS

- 1. Frequent and recurrent floods have had a significant impact on lives and livelihoods.** Bangladesh is a deltaic country consisting of floodplains created by over 400 rivers and channels, including three major rivers: Padma, the Brahmaputra-Jamuna, and the Meghna. Twenty five percent of the country is less than one meter above sea level, and 50 percent is less than 6 meters above sea level. It is located at the foot of the highest mountain range in the world, the Himalayas, which is also the world's highest precipitation zone. During the monsoon season, winds from the sea heading towards land raise the water levels in the Bay of Bengal, inhibiting drainage from the rivers into the sea. As rainfall levels are also high during the monsoon seasons, river flows, local rain, and raised water levels for the Bay of Bengal result in flooding of vast areas. Large scale floods in 2017 killed 145 people and affected about 8 million in 32 districts, putting 1.5 million people in need of immediate food assistance. Severe floods during the last week of June 2020, driven by prolonged and intensified heavy monsoon and upstream water, affected 5.4 million people in the northern, central, and north-eastern parts of the country. About 37 percent of the total area of the country was flooded affecting 33 districts; the disaster was the longest flooding period in the last 22 years.
- 2. Climate change increases lightning events by raising surface temperatures, which allows the air to hold more moisture, fueling more intense lightning events.** Higher temperatures enhance convective activity, leading to stronger updrafts and a higher likelihood of lightning.⁶⁶ Studies suggest that for every one degree increase in temperature, lightning activity could rise by 12 percent.⁶⁷ There is also a high mortality rate from lightning strikes, and according to the MoDMR, over 2,000 people died in lightning strikes in the country from 2011 to 2020⁶⁸.
- 3. Flooding and extreme precipitation, accelerated by climate change, continues to impact community infrastructure, including rural roads, bridges, flood control and drainage systems, and public buildings.** Floods in 2007 damaged over 14,000 kilometers of roads and washed away 849 bridges, 14 union council buildings and eight growth centers in 46 districts, with a rehabilitation cost of BDT 11,425.35 million (US\$134.4 million).⁶⁹ According to the BWDB, 87 kilometers of embankments were damaged in the affected districts in 2024. The BWDB also noted that 17 sections of flood control embankments along the Muhuri, Kahua, and Silonia rivers were breached on August 2, leading to further flooding. The Educational sector has been heavily impacted, with more than 7,000 schools closed during the 2024 floods, affecting 1.4 million children, including 749,390 girls in Chattogram Division alone.⁷⁰
- 4. Climate change has a significant impact on Bangladesh's river systems and floodplains, leading to increased flooding, riverbank erosion, and displacement of people, which results in substantial economic losses and challenges for the country's infrastructure and population.** Around 20–25 percent of the country floods during the monsoon every year. This number escalates up to 67 percent every four to five years in the face of a major flood event, induced from climate change. Concurrent riverbank erosion has been identified as a 'silent killer' as valuable land continues to be lost and about 100,000 people are displaced every year. The GoB estimates that the annual cost of

⁶⁶ Neal, N. (2021). How lightning is affected by climate change. *Prevention Web*. <https://www.preventionweb.net/news/how-lightning-affected-climate-change>.

⁶⁷ David M. Romps et al. 2014. "Projected increase in lightning strikes in the United States due to global warming." *Science* 346: 851-854. DOI: 10.1126/science.1259100

⁶⁸ GoB, MoDMR. 2020. *National Plan for Disaster Management (NPDM) 2021-2025*.

⁶⁹ GoB, Ministry of Food and Disaster Management, Disaster Management Bureau. 2007. *Consolidated Damage and Loss Assessment, Lessons Learnt from the Flood 2007 and Future Action Plan*.

⁷⁰ Inter-cluster Coordination Group (ICCG). 30 August, 2024. *Bangladesh: Eastern Flash Floods - Situation Report No. 02*. ReliefWeb.



climate change events could reach 1.5–3.0 percent of GDP by 2031,⁷¹ partly due to the inadequate climate adaptation of existing river infrastructure. The rapid acceleration of climate change is expected to intensify its impact on rivers and their floodplains. Unchecked erosion and sedimentation during heavy downpour, leading to wider, shallower rivers; and increased seasonal variability in water quantity due to intensified drought and heavy precipitation cycle. Erosion, sedimentation, and seasonal variability in water quantity are exacerbated by climate change.⁷²

5. **In the context of livelihoods, climate change will continue to affect agriculture, fisheries, and livestock sectors, making them more susceptible to the impacts of extreme weather.** Climate change is negatively impacting agricultural productivity in Bangladesh, with rising temperatures, unpredictable rainfall, and more frequent floods reducing the yield and profitability of farming activities. The predicted sea-level rise in the coastal zone is 0.2–1.0 meters in 2100, with a current trend of 6–20 millimeters year.⁷³ This poses additional risks to food security, including a projected decline of 6–9 percent in rice production this century.⁷⁴ Climate impact for livestock sector includes, in the short term, loss of animals by drowning, destruction of animal sheds, destruction of crops and infrastructure delivering services or inputs (veterinary posts, marketing infrastructure, hatcheries). Longer-term impacts include the likely emergence of climate sensitive diseases (mosquito-borne diseases, internal parasites such as fluke) as well as shortage of fodder and feed due to the destruction of crops.
6. **Poverty and vulnerability of the elderly widowed, and persons-with-disabilities are further aggravated by climate change.** Due to limited access to formal financial channels, the poor often tend to save in kind, which makes their assets more vulnerable to disaster shocks.⁷⁵ Frequent storm surges and consequential flooding and erosion impact a significant share of the already vulnerable populations. The combined effect of poverty and climate on vulnerable groups can be devastating. Weakened immune systems of older people make them more susceptible to diseases and less resilient in situations of water and food shortages which are inherent effects of climate change. Social isolation, poor mobility and limited access to resources and services further aggravate vulnerability. Women, especially when widowed in an essentially patriarchal society, and people with disabilities who represent approximately ten percent of the population, tend to be marginalized and lack agency and resources to garner resilience to climate shocks.

⁷¹ Gudmundsson, R., R. Moussa, and M. I. Hussain. 2019. "Climate Change Mitigation and Adaptation in Bangladesh: Policy Options." *IMF Country Report* No. 19/300, International Monetary Fund, Washington, DC.

⁷² Hossain, S., M. Rahman, F. Nusrat, R. Rahman, and N.F. Anisha. 2014. "Effects of Climate Change on River Morphology in Bangladesh and a Morphological Assessment of Sitalakha River." *Journal of River Research Institute* 1: 1-13.

⁷³ Mojid, M.A. 2020. "Climate change-induced challenges to sustainable development in Bangladesh." *IOP Conference Series: Earth and Environmental Science* 423(1), 012001.

⁷⁴ Ibid.

⁷⁵ Hallegatte, S. et al. 2016. *Shock waves: managing the impacts of climate change on poverty*. World Bank Publications.